

UPSC ISS · STATISTICS PAPER-I · 2027 FORECAST

Statistical Methods

FORECAST · AI-GENERATED · NOT PREVIOUS-YEAR QUESTIONS

Every forecast problem in this unit, with its verified answer and a ~30-second exam shortcut.

Forecast problems here	375 of 750 across the two units
Named forecast mocks	15, of exactly 25 questions each
Topics covered	13
Concepts covered	69
Layout of every entry	QUESTION → OPTIONS → ANSWER → EXAM SHORTCUT

These are not previous-year questions. Every stem, every option, every answer and every exam shortcut in this volume was written for this project as forecast practice for the 2027 attempt. Nothing here is reproduced from a UPSC paper and no official UPSC key exists for any of it.

Kept separate from the PYQ bank. The authentic 2018–2026 questions live in their own four volumes and in their own array inside the offline dashboard. Forecast material never enters a year, sectional, topic, subtopic or custom PYQ mock.

Answers were verified. Each key was worked out and independently re-checked during the build, but they remain a study aid, not an official key.

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Named forecast mocks

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FORECAST · STATISTICAL METHODS

Association of Attributes

21 forecast problems · 3 concepts

1. FORECAST · FS-039 Yule Coefficients

QUESTION

A two-way table has $a = 9, b = 1, c = 1$ and $d = 9$. What is Yule's coefficient of association Q ?

OPTIONS

- (a) 0.8
- (b) 40/41
- (c) 1
- (d) 0.9

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 40/41

EXAM SHORTCUT (AI-generated explanation)

$$Q = (ad - bc)/(ad + bc) = (81 - 1)/(81 + 1) = 40/41.$$

2. FORECAST · FS-040 Yule Coefficients

QUESTION

For the table $a = 9, b = 1, c = 1, d = 9$, what is Yule's coefficient of colligation Y ?

OPTIONS

- (a) 0.8
- (b) 40/41
- (c) 0.9
- (d) 0.64

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.8

EXAM SHORTCUT (AI-generated explanation)

$$Y = (\sqrt{ad} - \sqrt{bc})/(\sqrt{ad} + \sqrt{bc}) = (9 - 1)/(9 + 1) = 0.8.$$

3. FORECAST · FS-041 Yule Coefficients**QUESTION**

Which relation connects Yule's Q and Yule's Y?

OPTIONS

- (a) $Q = 2Y/(1 + Y^2)$
- (b) $Y = 2Q/(1 + Q^2)$
- (c) $Q = Y^2$
- (d) $Q = Y$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $Q = 2Y/(1 + Y^2)$

EXAM SHORTCUT (AI-generated explanation)

Writing $t = \sqrt{ad/bc}$, $Q = (t^2 - 1)/(t^2 + 1)$ and $Y = (t - 1)/(t + 1)$, which gives $Q = 2Y/(1 + Y^2)$.

4. FORECAST · FS-042 Yule Coefficients**QUESTION**

Which inequality always holds between Yule's Q and Yule's Y?

OPTIONS

- (a) $|Q| \leq |Y|$
- (b) $|Q| \geq |Y|$
- (c) $Q = -Y$
- (d) $Q + Y = 1$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $|Q| \geq |Y|$

EXAM SHORTCUT (AI-generated explanation)

$Q = 2Y/(1 + Y^2)$ and $1 + Y^2 \leq 2$ for $|Y| \leq 1$, so $|Q| \geq |Y|$.

5. FORECAST · FS-043 Yule Coefficients**QUESTION**

Yule's Q equals 1 if and only if

OPTIONS

- (a) $ad = 0$
- (b) $bc = 0$
- (c) $ad = bc$
- (d) $a = d$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $bc = 0$

EXAM SHORTCUT (AI-generated explanation)

$Q = (ad - bc)/(ad + bc)$ equals 1 exactly when $bc = 0$.

6. FORECAST · FS-044 Consistency of Data**QUESTION**

With $N = 100$, $(A) = 60$, $(B) = 70$ and $(AB) = 25$, the data are

OPTIONS

- (a) consistent
- (b) inconsistent, since a derived class frequency is negative
- (c) independent
- (d) positively associated

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) inconsistent, since a derived class frequency is negative

EXAM SHORTCUT (AI-generated explanation)

(AB) must be at least $60 + 70 - 100 = 30$, so 25 is impossible.

7. FORECAST · FS-045 Consistency of Data**QUESTION**

With $(A) = 40$, $(B) = 50$, $N = 200$ and $(AB) = 15$, the attributes are

OPTIONS

- (a) independent
- (b) positively associated
- (c) negatively associated
- (d) inconsistent

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) positively associated

EXAM SHORTCUT (AI-generated explanation)

The independence value is $40(50)/200 = 10$, and 15 exceeds it.

8. FORECAST · FS-046 Yule Coefficients**QUESTION**

The coefficient of colligation is

OPTIONS

- (a) Yule's Q
- (b) Yule's Y
- (c) Pearson's C
- (d) the ϕ coefficient

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) Yule's Y

EXAM SHORTCUT (AI-generated explanation)

Yule named Y the coefficient of colligation and Q the coefficient of association.

9. FORECAST · FS-047 Chi-Square and Contingency**QUESTION**

With a χ^2 value of 25 and $N = 100$, what is Pearson's coefficient of contingency C ?

OPTIONS

- (a) 0.5
- (b) 0.447
- (c) 0.25
- (d) 0.2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0.447

EXAM SHORTCUT (AI-generated explanation)

$$C = \sqrt{\chi^2 / (\chi^2 + N)} = \sqrt{25 / 125} = 0.447.$$

10. FORECAST · FS-048 Chi-Square and Contingency**QUESTION**

What is the maximum possible value of Pearson's C for a 2 by 2 table?

OPTIONS

- (a) 1
- (b) $1/\sqrt{2}$
- (c) $1/2$
- (d) $\sqrt{2}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/\sqrt{2}$

EXAM SHORTCUT (AI-generated explanation)

The maximum is $\sqrt{(k-1)/k}$ with $k = 2$, giving $1/\sqrt{2} = 0.707$.

11. **FORECAST · FS-049** Chi-Square and Contingency**QUESTION**

A two-way table has rows (30, 20) and (60, 40). What is the χ^2 statistic?

OPTIONS

- (a) 0
- (b) 1
- (c) 2.5
- (d) 10

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0

EXAM SHORTCUT (AI-generated explanation)

The rows are exactly proportional, so $ad = bc$ and χ^2 is 0.

12. **FORECAST · FS-050** Chi-Square and Contingency**QUESTION**

For the table $a = 10, b = 20, c = 20, d = 10$, what is the χ^2 statistic?

OPTIONS

- (a) $20/3$
- (b) 10
- (c) 5
- (d) 3

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $20/3$

EXAM SHORTCUT (AI-generated explanation)

$N(ad - bc)^2$ divided by the four marginal products: $60(300)^2/(30^4) = 20/3$.

13. **FORECAST · FS-051** Yule Coefficients**QUESTION**

If all four cell frequencies of a 2 by 2 table are equal, Yule's Q is

OPTIONS

- (a) 1
- (b) -1
- (c) 0
- (d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 0

EXAM SHORTCUT (AI-generated explanation)

$ad = bc$, so the numerator of Q vanishes.

14. **FORECAST · FS-052** Yule Coefficients**QUESTION**

Yule's Q is invariant under

OPTIONS

- (a) multiplying a whole row by a constant
- (b) adding a constant to every cell
- (c) interchanging a and b
- (d) none of these

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) multiplying a whole row by a constant

EXAM SHORTCUT (AI-generated explanation)

Q depends only on the ratio ad/bc , and scaling a row multiplies both products equally.

15. **FORECAST · FS-053** Consistency of Data**QUESTION**

For three attributes, the total number of class frequencies of all orders is

OPTIONS

- (a) 8
- (b) 9
- (c) 27
- (d) 6

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 27

EXAM SHORTCUT (AI-generated explanation)

Each attribute is present, absent or ignored, giving $3^3 = 27$ class frequencies.

16. **FORECAST · FS-054** Consistency of Data**QUESTION**

With $(A) = 50$, $(B) = 60$ and $N = 100$, what is the admissible range of (AB) ?

OPTIONS

- (a) 0 to 50
- (b) 10 to 50
- (c) 10 to 60
- (d) 0 to 60

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 10 to 50

EXAM SHORTCUT (AI-generated explanation)

The lower bound is $(A) + (B) - N = 10$ and the upper bound is $\min(50, 60) = 50$.

17. FORECAST · FS-055 Consistency of Data**QUESTION**

If $(AB)/(A) = 0.6$ while $(B)/N = 0.5$, the attributes are

OPTIONS

- (a) independent
- (b) positively associated
- (c) negatively associated
- (d) inconsistent

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) positively associated

EXAM SHORTCUT (AI-generated explanation)

The conditional proportion 0.6 exceeds the unconditional 0.5, so A raises the chance of B.

18. FORECAST · FS-056 Chi-Square and Contingency**QUESTION**

For a 2 by 2 table, the ϕ coefficient equals

OPTIONS

- (a) the square root of χ^2 divided by N
- (b) χ^2 divided by N
- (c) the square root of χ^2 divided by $(\chi^2 + N)$
- (d) χ^2 divided by $(N - 1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) the square root of χ^2 divided by N

EXAM SHORTCUT (AI-generated explanation)

$\phi = \sqrt{\chi^2/N}$, which for a 2 by 2 table equals the Pearson correlation of the two indicators.

19. **FORECAST · FS-346** Yule Coefficients**QUESTION**

If Yule's Y equals 0.5, what is Yule's Q ?

OPTIONS

- (a) 0.8
- (b) 0.5
- (c) 0.25
- (d) 0.9

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.8

EXAM SHORTCUT (AI-generated explanation)

$$Q = 2Y/(1 + Y^2) = 1/1.25 = 0.8.$$

20. **FORECAST · FS-347** Chi-Square and Contingency**QUESTION**

For the table $a = 10, b = 5, c = 5, d = 10$, what is the Yates-corrected χ^2 ?

OPTIONS

- (a) 2.13
- (b) 3.33
- (c) 6.67
- (d) 1.2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 2.13

EXAM SHORTCUT (AI-generated explanation)

$$30(|100 - 25| - 15)^2/(15^4) = 30(3600)/50625 = 2.133.$$

21. **FORECAST · FS-367** Chi-Square and Contingency**QUESTION**

In a χ^2 test of independence, the expected frequency of a cell is

OPTIONS

- (a) the row total times the column total divided by the grand total
- (b) the row total plus the column total
- (c) the grand total divided by the number of cells
- (d) the observed frequency

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) the row total times the column total divided by the grand total

EXAM SHORTCUT (AI-generated explanation)

Independence makes the joint proportion the product of the marginal proportions.

FORECAST · STATISTICAL METHODS

Asymptotic Relative Efficiency

5 forecast problems · 1 concept

1. FORECAST · FS-320 Efficiency Comparisons**QUESTION**

For a normal population, the asymptotic relative efficiency of the sample median relative to the sample mean is

OPTIONS

- (a) $\pi/2$
- (b) $2/\pi$
- (c) 1
- (d) $3/\pi$

ANSWER (AI-generated forecast item — no official UPSC key exists)**(b)** $2/\pi$ **EXAM SHORTCUT** (AI-generated explanation)

The median has asymptotic variance $\pi\sigma^2/(2n)$ against σ^2/n for the mean, giving $2/\pi = 0.637$.

2. FORECAST · FS-321 Efficiency Comparisons**QUESTION**

For $N(\mu, \sigma^2)$, the sample mean beats the sample median

OPTIONS

- (a) only if $\sigma^2 < 1$
- (b) only if $\mu = 0$
- (c) for every μ and σ^2
- (d) only for large n

ANSWER (AI-generated forecast item — no official UPSC key exists)**(c)** for every μ and σ^2 **EXAM SHORTCUT** (AI-generated explanation)

The efficiency ratio $2/\pi$ is free of both parameters, so the mean always wins under normality.

3. FORECAST · FS-322 Efficiency Comparisons**QUESTION**

For the Laplace (double exponential) distribution, the asymptotic relative efficiency of the median relative to the mean is

OPTIONS

- (a) $1/2$
- (b) 2
- (c) 1
- (d) $\pi/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 2

EXAM SHORTCUT (AI-generated explanation)

The Laplace density peaks sharply at the median, making it twice as efficient as the mean.

4. FORECAST · FS-323 Efficiency Comparisons**QUESTION**

For a Cauchy population, the sample mean is

OPTIONS

- (a) consistent
- (b) not consistent, whereas the median is
- (c) efficient
- (d) unbiased with a finite variance

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) not consistent, whereas the median is

EXAM SHORTCUT (AI-generated explanation)

The Cauchy sample mean has the same Cauchy law as a single observation, while the median is consistent.

5. FORECAST · FS-324 Efficiency Comparisons**QUESTION**

The asymptotic relative efficiency of estimator 1 relative to estimator 2 equals

OPTIONS

- (a) the asymptotic variance of 1 divided by that of 2
- (b) the asymptotic variance of 2 divided by that of 1
- (c) the ratio of their variances at $n = 1$
- (d) the ratio of their biases

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the asymptotic variance of 2 divided by that of 1

EXAM SHORTCUT (AI-generated explanation)

Efficiency is inversely proportional to variance, so the comparison estimator's variance goes on top.

FORECAST · STATISTICAL METHODS

Bivariate Normal Distribution

44 forecast problems · 8 concepts

1. FORECAST · FS-064 Moments of the Bivariate Normal**QUESTION**

For the standard bivariate normal with correlation ρ , what is $E(X^2Y^2)$?

OPTIONS

- (a) 1
- (b) $1 + \rho^2$
- (c) $1 + 2\rho^2$
- (d) $2\rho^2$

ANSWER (AI-generated forecast item — no official UPSC key exists)**(c)** $1 + 2\rho^2$ **EXAM SHORTCUT** (AI-generated explanation)

Isserlis' theorem gives $E(X^2Y^2) = E(X^2)E(Y^2) + 2[E(XY)]^2 = 1 + 2\rho^2$.

2. FORECAST · FS-065 Conditional Distributions**QUESTION**

In a bivariate normal with $\sigma_2 = 4$ and $\rho = 0.6$, what is $\text{Var}(Y \text{ given } X = x)$?

OPTIONS

- (a) 6.4
- (b) 10.24
- (c) 16
- (d) 9.6

ANSWER (AI-generated forecast item — no official UPSC key exists)**(b)** 10.24**EXAM SHORTCUT** (AI-generated explanation)

$16(1 - 0.36) = 10.24$.

3. FORECAST · FS-066 Exponent Matching**QUESTION**

A bivariate density is $f(x, y) = \lambda \exp[-(x^2 - xy + y^2)]$. What is $\text{Var}(X)$?

OPTIONS

- (a) 1
- (b) $2/3$
- (c) $1/2$
- (d) $3/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $2/3$

EXAM SHORTCUT (AI-generated explanation)

Matching the quadratic form gives $\sigma^2 = 2/3$ with $\rho = 1/2$.

4. FORECAST · FS-067 Exponent Matching**QUESTION**

For $f(x, y) = \lambda \exp[-(x^2 - xy + y^2)]$, what is ρ ?

OPTIONS

- (a) $1/2$
- (b) $-1/2$
- (c) $1/4$
- (d) 0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/2$

EXAM SHORTCUT (AI-generated explanation)

The ratio of the xy coefficient to the x^2 coefficient gives $2\rho = 1$.

5. FORECAST · FS-068 Exponent Matching**QUESTION**

For $f(x, y) = \lambda \exp[-(x^2 - xy + y^2)]$, what is λ ?

OPTIONS

- (a) $1/(2\pi)$
- (b) $\sqrt{3}/(2\pi)$
- (c) $1/(\pi\sqrt{3})$
- (d) $3/(2\pi)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\sqrt{3}/(2\pi)$

EXAM SHORTCUT (AI-generated explanation)

$$\lambda = 1/(2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}) = 1/(2\pi(2/3)\sqrt{3}/2) = \sqrt{3}/(2\pi).$$

6. FORECAST · FS-069 Regression Lines**QUESTION**

For $f(x, y) = \lambda \exp[-(x^2 - xy + y^2)]$, the regression of Y on X is

OPTIONS

- (a) $y = x$
- (b) $y = x/2$
- (c) $y = 2x$
- (d) $y = -x/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $y = x/2$

EXAM SHORTCUT (AI-generated explanation)

The slope is $\rho\sigma_2/\sigma_1 = 1/2$ since the two variances are equal.

7. FORECAST · FS-070 Exponent Matching**QUESTION**

For $f(x, y) = \lambda \exp[-(2/3)(x^2 - xy + y^2)]$, what is λ ?

OPTIONS

- (a) $1/(\pi\sqrt{3})$
- (b) $\sqrt{3}/(2\pi)$
- (c) $1/(2\pi)$
- (d) $1/\pi$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/(\pi\sqrt{3})$

EXAM SHORTCUT (AI-generated explanation)

Here $\rho = 1/2$ again and $\sigma^2 = 1$, so $\lambda = 1/(2\pi\sqrt{3}/2) = 1/(\pi\sqrt{3})$.

8. FORECAST · FS-071 Exponent Matching**QUESTION**

For $f(x, y) = \lambda \exp[-(4x^2 + y^2)/2]$, what is λ ?

OPTIONS

- (a) $1/(2\pi)$
- (b) $1/\pi$
- (c) $2/\pi$
- (d) $1/(4\pi)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/\pi$

EXAM SHORTCUT (AI-generated explanation)

No cross term means $\rho = 0$ with $\sigma_1^2 = 1/4$ and $\sigma_2^2 = 1$, so $\lambda = 1/(2\pi(1/2)(1)) = 1/\pi$.

9. FORECAST · FS-072 Marginal Distributions**QUESTION**

In a bivariate normal with parameters $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2$ and ρ , the marginal of X is

OPTIONS

- (a) $N(\mu_1, \sigma_1^2(1 - \rho^2))$
- (b) $N(\mu_1, \sigma_1^2)$
- (c) $N(\mu_1 + \rho\sigma_1, \sigma_1^2)$
- (d) not normal

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $N(\mu_1, \sigma_1^2)$

EXAM SHORTCUT (AI-generated explanation)

Marginals of a bivariate normal keep the full variance; only CONDITIONAL variances are shrunk by $1 - \rho^2$.

10. FORECAST · FS-073 Independence and Linear Combinations**QUESTION**

For a bivariate normal, $X + Y$ and $X - Y$ are independent if and only if

OPTIONS

- (a) $\rho = 0$
- (b) $\sigma_1 = \sigma_2$
- (c) $\mu_1 = \mu_2$
- (d) always

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\sigma_1 = \sigma_2$

EXAM SHORTCUT (AI-generated explanation)

$\text{Cov}(X + Y, X - Y) = \sigma_1^2 - \sigma_2^2$, which vanishes exactly when the variances are equal.

11. **FORECAST · FS-074** Marginal Distributions**QUESTION**

If X and Y each have normal marginals, then (X, Y) is bivariate normal

OPTIONS

- (a) always
- (b) never
- (c) not necessarily
- (d) only if $\rho = 0$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) not necessarily

EXAM SHORTCUT (AI-generated explanation)

Normal marginals are far weaker than joint normality; standard counterexamples exist.

12. **FORECAST · FS-075** Independence and Linear Combinations**QUESTION**

A pair (X, Y) is bivariate normal if and only if

OPTIONS

- (a) X and Y are each normal
- (b) $aX + bY$ is normal for every a and b
- (c) $X + Y$ is normal
- (d) XY is normal

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $aX + bY$ is normal for every a and b

EXAM SHORTCUT (AI-generated explanation)

Joint normality is equivalent to normality of every linear combination - the Cramer-Wold device.

13. FORECAST · FS-076 Independence and Linear Combinations**QUESTION**

Zero correlation implies independence

OPTIONS

- (a) for any bivariate distribution
- (b) only for the bivariate normal
- (c) never
- (d) only when $\sigma_1 = \sigma_2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) only for the bivariate normal

EXAM SHORTCUT (AI-generated explanation)

The implication is a special property of joint normality, not a general fact.

14. FORECAST · FS-077 Moments of the Bivariate Normal**QUESTION**

For the standard bivariate normal with correlation ρ , what is $E(e^{X+Y})$?

OPTIONS

- (a) e
- (b) $e^{1+\rho}$
- (c) $e^{2+2\rho}$
- (d) e^ρ

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $e^{1+\rho}$

EXAM SHORTCUT (AI-generated explanation)

$X + Y$ is $N(0, 2 + 2\rho)$, so the MGF at 1 is $\exp((2 + 2\rho)/2) = e^{1+\rho}$.

15. FORECAST · FS-078 Independence and Linear Combinations**QUESTION**

With $\sigma_1 = 3$, $\sigma_2 = 4$ and $\rho = 0.5$, what is $\text{Var}(X + Y)$?

OPTIONS

- (a) 25
- (b) 37
- (c) 13
- (d) 31

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 37

EXAM SHORTCUT (AI-generated explanation)

$$9 + 16 + 2(0.5)(3)(4) = 25 + 12 = 37.$$

16. FORECAST · FS-079 Independence and Linear Combinations**QUESTION**

For the standard bivariate normal with correlation ρ , what is $\text{Var}(2X - 3Y)$?

OPTIONS

- (a) 13
- (b) $13 - 12\rho$
- (c) $13 + 12\rho$
- (d) 5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $13 - 12\rho$

EXAM SHORTCUT (AI-generated explanation)

$$4 + 9 - 2(2)(3)\rho = 13 - 12\rho.$$

17. **FORECAST · FS-080** Regression Lines**QUESTION**

For the standard bivariate normal, what is $\text{Cov}(Y - \rho X, X)$?

OPTIONS

- (a) ρ
- (b) $1 - \rho^2$
- (c) 0
- (d) ρ^2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 0

EXAM SHORTCUT (AI-generated explanation)

$$\text{Cov}(Y, X) - \rho \text{Var}(X) = \rho - \rho = 0.$$

18. **FORECAST · FS-081** Regression Lines**QUESTION**

For a general bivariate normal, what is $\text{Cov}(Y - b_{yx}X, X)$, where b_{yx} is the regression coefficient of Y on X?

OPTIONS

- (a) 0
- (b) $\rho\sigma_1\sigma_2$
- (c) $\sigma_2^2(1 - \rho^2)$
- (d) $\rho\sigma_1^2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0

EXAM SHORTCUT (AI-generated explanation)

By construction the regression residual is uncorrelated with the predictor.

19. **FORECAST · FS-082** Conditional Distributions**QUESTION**

With $\mu_1 = 10, \mu_2 = 20, \sigma_1 = 2, \sigma_2 = 5$ and $\rho = 0.8$, what is $E(Y \text{ given } X = 12)$?

OPTIONS

- (a) 22
- (b) 24
- (c) 23
- (d) 25

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 24

EXAM SHORTCUT (AI-generated explanation)

$$20 + 0.8(5/2)(2) = 20 + 4 = 24.$$

20. **FORECAST · FS-083** Conditional Distributions**QUESTION**

With $\sigma_2 = 5$ and $\rho = 0.8$, what is $\text{Var}(Y \text{ given } X)$?

OPTIONS

- (a) 9
- (b) 16
- (c) 25
- (d) 4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 9

EXAM SHORTCUT (AI-generated explanation)

$$25(1 - 0.64) = 9.$$

21. **FORECAST · FS-084** Conditional Distributions**QUESTION**

With $\mu_1 = 10, \mu_2 = 20, \sigma_1 = 2, \sigma_2 = 5, \rho = 0.8$, what is $P(Y > 20 \text{ given } X = 12)$ approximately?

OPTIONS

- (a) 0.09
- (b) 0.50
- (c) 0.91
- (d) 0.68

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 0.91

EXAM SHORTCUT (AI-generated explanation)

The conditional law is $N(24, 9)$, so $z = (20 - 24)/3 = -1.33$ and the upper tail is about 0.91.

22. **FORECAST · FS-085** Quadratic Forms**QUESTION**

For the standard bivariate normal with correlation ρ , the quantity $(X^2 + Y^2 - 2\rho XY)/(1 - \rho^2)$ follows

OPTIONS

- (a) χ^2 with 1 degree of freedom
- (b) χ^2 with 2 degrees of freedom
- (c) $N(0, 1)$
- (d) $F(1, 1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) χ^2 with 2 degrees of freedom

EXAM SHORTCUT (AI-generated explanation)

This is the Mahalanobis quadratic form of a bivariate normal, which is χ^2 with 2 degrees of freedom.

23. **FORECAST · FS-086** Quadratic Forms**QUESTION**

For the standard bivariate normal, what is the MGF of $X^2 + Y^2 - 2\rho XY$?

OPTIONS

- (a) $(1 - 2t)^{-1}$
- (b) $[1 - 2(1 - \rho^2)t]^{-1}$
- (c) $(1 - 2t)^{-2}$
- (d) $e^{t(1-\rho^2)}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $[1 - 2(1 - \rho^2)t]^{-1}$

EXAM SHORTCUT (AI-generated explanation)

The quantity is $(1 - \rho^2)$ times a χ^2 with 2 degrees of freedom, whose MGF is $(1 - 2t)^{-1}$.

24. **FORECAST · FS-087** Quadratic Forms**QUESTION**

For an independent standard bivariate normal pair, $X^2 + Y^2$ follows

OPTIONS

- (a) an exponential with mean 1
- (b) an exponential with mean 2
- (c) χ^2 with 1 degree of freedom
- (d) $N(2, 4)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) an exponential with mean 2

EXAM SHORTCUT (AI-generated explanation)

χ^2 with 2 degrees of freedom is exactly the exponential with mean 2.

25. **FORECAST · FS-088** Regression Lines**QUESTION**

The two regression lines are perpendicular when

OPTIONS

- (a) $r = \text{plus or minus } 1$
- (b) $r = 0$
- (c) $\sigma_1 = \sigma_2$
- (d) never

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $r = 0$

EXAM SHORTCUT (AI-generated explanation)

At $r = 0$ the lines become $y = \bar{y}$ and $x = \bar{x}$, which are at right angles.

26. **FORECAST · FS-089** Regression Lines**QUESTION**

The two regression lines coincide when

OPTIONS

- (a) $r = 0$
- (b) $r = \text{plus or minus } 1$
- (c) $\mu_1 = \mu_2$
- (d) $\sigma_1 = \sigma_2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $r = \text{plus or minus } 1$

EXAM SHORTCUT (AI-generated explanation)

Perfect correlation makes both lines the same exact linear relation.

27. FORECAST · FS-090 Conditional Distributions**QUESTION**

In a bivariate normal, $\text{Var}(Y \text{ given } X = x)$ as a function of x is

OPTIONS

- (a) increasing in x
- (b) decreasing in x
- (c) constant in x
- (d) dependent on μ_1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) constant in x

EXAM SHORTCUT (AI-generated explanation)

The conditional variance $\sigma_2^2(1 - \rho^2)$ does not involve x at all.

28. FORECAST · FS-091 Independence and Linear Combinations**QUESTION**

X and Y are independent $N(0, 1)$. What is $\text{Corr}(X, X + Y)$?

OPTIONS

- (a) $1/2$
- (b) $1/\sqrt{2}$
- (c) 1
- (d) 0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/\sqrt{2}$

EXAM SHORTCUT (AI-generated explanation)

$\text{Cov} = 1$ and $\text{Var}(X + Y) = 2$, so the correlation is $1/\sqrt{2}$.

29. FORECAST · FS-092 Independence and Linear Combinations**QUESTION**

With $\rho = 0$ but σ_1 not equal to σ_2 , what is $\text{Corr}(X + Y, X - Y)$?

OPTIONS

- (a) 0
- (b) $(\sigma_1^2 - \sigma_2^2)/(\sigma_1^2 + \sigma_2^2)$
- (c) 1
- (d) $\sigma_1\sigma_2/(\sigma_1^2 + \sigma_2^2)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $(\sigma_1^2 - \sigma_2^2)/(\sigma_1^2 + \sigma_2^2)$

EXAM SHORTCUT (AI-generated explanation)

$\text{Cov} = \sigma_1^2 - \sigma_2^2$ and both variances are $\sigma_1^2 + \sigma_2^2$.

30. FORECAST · FS-093 Moments of the Bivariate Normal**QUESTION**

For a general bivariate normal, $E(XY)$ equals

OPTIONS

- (a) $\mu_1\mu_2$
- (b) $\rho\sigma_1\sigma_2$
- (c) $\mu_1\mu_2 + \rho\sigma_1\sigma_2$
- (d) ρ

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $\mu_1\mu_2 + \rho\sigma_1\sigma_2$

EXAM SHORTCUT (AI-generated explanation)

$E(XY) = \text{Cov}(X, Y) + E(X)E(Y)$.

31. **FORECAST · FS-094** Orthant Probabilities**QUESTION**

For the standard bivariate normal with $\rho = 1/2$, what is $P(X > 0 \text{ and } Y > 0)$?

OPTIONS

- (a) $1/4$
- (b) $1/3$
- (c) $1/2$
- (d) $1/6$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/3$

EXAM SHORTCUT (AI-generated explanation)

The orthant formula $1/4 + \arcsin(\rho)/(2\pi)$ gives $1/4 + (\pi/6)/(2\pi) = 1/3$.

32. **FORECAST · FS-095** Orthant Probabilities**QUESTION**

For the standard bivariate normal with $\rho = 0$, what is $P(X > 0 \text{ and } Y > 0)$?

OPTIONS

- (a) $1/4$
- (b) $1/2$
- (c) $1/3$
- (d) 0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/4$

EXAM SHORTCUT (AI-generated explanation)

Independence gives $(1/2)(1/2) = 1/4$, which is also the orthant formula with $\arcsin(0) = 0$.

33. **FORECAST · FS-096** Regression Lines**QUESTION**

The lines $4x - 5y + 33 = 0$ and $20x - 9y = 107$ are the two regression lines. What is the point of means?

OPTIONS

- (a) (13, 17)
- (b) (17, 13)
- (c) (11, 15)
- (d) (12, 16)

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) (13, 17)

EXAM SHORTCUT (AI-generated explanation)

Solving the two equations simultaneously gives $x = 13$ and $y = 17$.

34. **FORECAST · FS-097** Regression Lines**QUESTION**

For the regression lines $4x - 5y + 33 = 0$ and $20x - 9y = 107$, what is r ?

OPTIONS

- (a) 0.6
- (b) 0.36
- (c) 0.8
- (d) -0.6

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.6

EXAM SHORTCUT (AI-generated explanation)

Taking the first as Y on X gives $b_{yx} = 4/5$ and the second as X on Y gives $b_{xy} = 9/20$; the product 0.36 gives $r = 0.6$.

35. **FORECAST · FS-098** Regression Lines**QUESTION**

For the regression lines $4x - 5y + 33 = 0$ and $20x - 9y = 107$ with the variance of X equal to 9, what is the standard deviation of Y ?

OPTIONS

- (a) 3
- (b) 4
- (c) 5
- (d) 2.4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 4

EXAM SHORTCUT (AI-generated explanation)

$$b_{yx} = r\sigma_y/\sigma_x \text{ gives } 0.8 = 0.6\sigma_y/3, \text{ so } \sigma_y = 4.$$

36. **FORECAST · FS-099** Conditional Distributions**QUESTION**

For the standard bivariate normal, what is $E(X^2 \text{ given } Y = y)$?

OPTIONS

- (a) $\rho^2 y^2$
- (b) $1 - \rho^2 + \rho^2 y^2$
- (c) 1
- (d) ρy

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1 - \rho^2 + \rho^2 y^2$

EXAM SHORTCUT (AI-generated explanation)

$$E(X^2 | Y) = \text{Var}(X | Y) + [E(X | Y)]^2 = (1 - \rho^2) + (\rho y)^2.$$

37. **FORECAST · FS-100** Moments of the Bivariate Normal**QUESTION**

For the standard bivariate normal, what is $\text{Corr}(X^2, Y^2)$?

OPTIONS

- (a) ρ
- (b) ρ^2
- (c) $2\rho^2$
- (d) $1 + 2\rho^2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) ρ^2

EXAM SHORTCUT (AI-generated explanation)

$\text{Cov}(X^2, Y^2) = 2\rho^2$ and $\text{Var}(X^2) = 2$, so the correlation is ρ^2 .

38. **FORECAST · FS-101** Regression Lines**QUESTION**

In a bivariate normal, X and the residual $Y - E(Y \text{ given } X)$ are

OPTIONS

- (a) uncorrelated but dependent
- (b) independent
- (c) perfectly correlated
- (d) identically distributed

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) independent

EXAM SHORTCUT (AI-generated explanation)

The residual is a linear combination of jointly normal variables and is uncorrelated with X , hence independent of it.

39. FORECAST · FS-102 Independence and Linear Combinations**QUESTION**

X is $N(0, 1)$ and Z is $N(0, 3)$, independent, and $Y = X + Z$. What is $\text{Corr}(X, Y)$?

OPTIONS

- (a) $1/4$
- (b) $1/2$
- (c) $1/\sqrt{3}$
- (d) $1/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/2$

EXAM SHORTCUT (AI-generated explanation)

$\text{Cov} = 1$ and $\text{Var}(Y) = 4$, so the correlation is $1/2$.

40. FORECAST · FS-103 Marginal Distributions**QUESTION**

If ρ equals plus or minus 1 in a bivariate normal, then

OPTIONS

- (a) the joint density still exists
- (b) the distribution is degenerate and has no joint density
- (c) X and Y are independent
- (d) ρ is undefined

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the distribution is degenerate and has no joint density

EXAM SHORTCUT (AI-generated explanation)

The covariance matrix becomes singular, so the mass concentrates on a line and no two-dimensional density exists.

41. FORECAST · FS-326 Moments of the Bivariate Normal**QUESTION**

For the standard bivariate normal with correlation ρ , what is $\text{Var}(XY)$?

OPTIONS

- (a) $1 + \rho^2$
- (b) $1 + 2\rho^2$
- (c) ρ^2
- (d) 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1 + \rho^2$

EXAM SHORTCUT (AI-generated explanation)

$E(X^2Y^2) = 1 + 2\rho^2$ and $E(XY) = \rho$, so the variance is $1 + 2\rho^2 - \rho^2$.

42. FORECAST · FS-327 Conditional Distributions**QUESTION**

For the standard bivariate normal with correlation ρ , the conditional law of X given $X + Y = s$ is

OPTIONS

- (a) $N(s/2, (1 - \rho)/2)$
- (b) $N(s/2, 1 - \rho^2)$
- (c) $N(\rho s, 1 - \rho^2)$
- (d) $N(s, 1 + \rho)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $N(s/2, (1 - \rho)/2)$

EXAM SHORTCUT (AI-generated explanation)

The regression slope on the sum is $(1 + \rho)/(2 + 2\rho) = 1/2$ and the residual variance is $1 - (1 + \rho)/2$.

43. **FORECAST · FS-357** Orthant Probabilities**QUESTION**

For the standard bivariate normal with $\rho = 1/2$, what is $P(X > 0 \text{ and } Y < 0)$?

OPTIONS

- (a) $1/4$
- (b) $1/6$
- (c) $1/3$
- (d) $1/12$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/6$

EXAM SHORTCUT (AI-generated explanation)

$$1/4 - \arcsin(\rho)/(2\pi) = 1/4 - 1/12 = 1/6.$$

44. **FORECAST · FS-362** Regression Lines**QUESTION**

In a bivariate normal with $b_{yx} = 0.8$ and $\rho = 0.4$, what is the ratio of σ_2 to σ_1 ?

OPTIONS

- (a) 2
- (b) 0.5
- (c) 0.32
- (d) 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 2

EXAM SHORTCUT (AI-generated explanation)

$$b_{yx} = \rho\sigma_2/\sigma_1 \text{ gives } 0.8 = 0.4(\text{ratio}), \text{ so the ratio is } 2.$$

FORECAST · STATISTICAL METHODS

Correlation Analysis

69 forecast problems · 12 concepts

1. FORECAST · FS-122 Invariance of Correlation**QUESTION**

What is the correlation between $X + 3$ and Y ?

OPTIONS

- (a) $r + 3$
- (b) r
- (c) $3r$
- (d) $r/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)**(b)** r **EXAM SHORTCUT** (AI-generated explanation)

A shift changes neither the covariance nor either variance, so r is unchanged.

2. FORECAST · FS-123 Invariance of Correlation**QUESTION**

What is the correlation between X and $-2Y + 5$?

OPTIONS

- (a) $-2r$
- (b) $-r$
- (c) r
- (d) $2r$

ANSWER (AI-generated forecast item — no official UPSC key exists)**(b)** $-r$ **EXAM SHORTCUT** (AI-generated explanation)

The negative multiplier flips the sign; its magnitude cancels in the standardisation.

3. FORECAST · FS-124 Invariance of Correlation**QUESTION**

What is the correlation between $4x - 3$ and $(7 - 4y)/8$?

OPTIONS

- (a) r
- (b) $-r$
- (c) $-r/2$
- (d) $2r$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $-r$

EXAM SHORTCUT (AI-generated explanation)

One positive and one negative multiplier give an overall sign change and no change in magnitude.

4. FORECAST · FS-125 Fisher Z Transformation**QUESTION**

With $n = 28$, what is the standard error of Fisher's Z?

OPTIONS

- (a) 0.2
- (b) 0.19
- (c) 0.04
- (d) 0.25

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.2

EXAM SHORTCUT (AI-generated explanation)

$SE(Z) = 1/\sqrt{n-3} = 1/5 = 0.2.$

5. FORECAST · FS-126 Fisher Z Transformation**QUESTION**

With $r = 0.6$, what is Fisher's Z?

OPTIONS

- (a) $\log 2$
- (b) $\log 4$
- (c) 0.6
- (d) half the log of 1.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\log 2$

EXAM SHORTCUT (AI-generated explanation)

$$Z = (1/2) \log((1+r)/(1-r)) = (1/2) \log 4 = \log 2.$$

6. FORECAST · FS-127 Testing a Correlation**QUESTION**

With $r = 0.6$ and $n = 27$, what is the t statistic for testing that ρ is zero?

OPTIONS

- (a) 3.75
- (b) 3
- (c) 2.5
- (d) 4.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 3.75

EXAM SHORTCUT (AI-generated explanation)

$$t = r\sqrt{n-2}/\sqrt{1-r^2} = 0.6(5)/0.8 = 3.75.$$

7. FORECAST · FS-128 Testing a Correlation**QUESTION**

With $n = 27$ and a critical t value of 2.06 on 25 degrees of freedom, the smallest $|r|$ significant at the 5% level is about

OPTIONS

- (a) 0.28
- (b) 0.38
- (c) 0.48
- (d) 0.58

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0.38

EXAM SHORTCUT (AI-generated explanation)

Solving $5r = 2.06\sqrt{1 - r^2}$ gives $r^2 = 4.2436/29.2436$, so $r = 0.381$.

8. FORECAST · FS-129 Partial Correlation**QUESTION**

With $r_{12} = 0.8$, $r_{13} = 0.6$ and $r_{23} = 0.5$, what is $r_{12.3}$ approximately?

OPTIONS

- (a) 0.72
- (b) 0.50
- (c) 0.65
- (d) 0.80

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.72

EXAM SHORTCUT (AI-generated explanation)

$(0.8 - 0.3)/\sqrt{0.64 \times 0.75} = 0.5/0.6928 = 0.722$.

9. FORECAST · FS-130 Partial Correlation**QUESTION**

If all three pairwise correlations equal 0.5, what is $r_{12.3}$?

OPTIONS

- (a) $1/3$
- (b) $1/2$
- (c) $1/4$
- (d) $2/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/3$

EXAM SHORTCUT (AI-generated explanation)

$$(0.5 - 0.25)/(1 - 0.25) = 0.25/0.75 = 1/3.$$

10. FORECAST · FS-131 Partial Correlation**QUESTION**

If all three pairwise correlations equal r , then $r_{12.3}$ equals

OPTIONS

- (a) $r/(1 + r)$
- (b) $r/(1 - r)$
- (c) r^2
- (d) $r/(1 + r^2)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $r/(1 + r)$

EXAM SHORTCUT (AI-generated explanation)

$$(r - r^2)/(1 - r^2) = r(1 - r)/((1 - r)(1 + r)) = r/(1 + r).$$

11. **FORECAST · FS-132** Multiple Correlation**QUESTION**

If all three pairwise correlations equal 0.5, what is the squared multiple correlation of variable 1 on variables 2 and 3?

OPTIONS

- (a) $1/3$
- (b) $1/2$
- (c) $2/3$
- (d) $1/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/3$

EXAM SHORTCUT (AI-generated explanation)

$$(0.25 + 0.25 - 2(0.125))/(1 - 0.25) = 0.25/0.75 = 1/3.$$

12. **FORECAST · FS-133** Multiple Correlation**QUESTION**

If all three pairwise correlations equal r , the squared multiple correlation of variable 1 on 2 and 3 is

OPTIONS

- (a) $2r^2/(1 + r)$
- (b) $2r/(1 + r)$
- (c) $r^2/(1 + r)$
- (d) $2r^2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $2r^2/(1 + r)$

EXAM SHORTCUT (AI-generated explanation)

$$(2r^2 - 2r^3)/(1 - r^2) = 2r^2(1 - r)/((1 - r)(1 + r)) = 2r^2/(1 + r).$$

13. **FORECAST · FS-134** Multiple Correlation**QUESTION**

Which relation always holds for the squared multiple correlation of variable 1 on variables 2 and 3?

OPTIONS

- (a) it is at most the smaller of r_{12}^2 and r_{13}^2
- (b) it is at least the larger of r_{12}^2 and r_{13}^2
- (c) it equals $r_{12}^2 + r_{13}^2$
- (d) it is at most r_{12}^2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) it is at least the larger of r_{12}^2 and r_{13}^2

EXAM SHORTCUT (AI-generated explanation)

Adding a predictor can never reduce the explained variation, so the multiple R^2 dominates each simple r^2 .

14. **FORECAST · FS-135** Multiple Correlation**QUESTION**

If the multiple correlation of variable 1 on variables 2 and 3 is zero, then

OPTIONS

- (a) $r_{23} = 0$
- (b) $r_{12} = r_{13} = 0$
- (c) $r_{12} = r_{13}$
- (d) nothing follows

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $r_{12} = r_{13} = 0$

EXAM SHORTCUT (AI-generated explanation)

Zero explained variation forces both simple correlations to vanish.

15. FORECAST · FS-136 Intraclass Correlation**QUESTION**

For an intraclass correlation with $k = 5$ members per family, the admissible range is

OPTIONS

- (a) -1 to 1
- (b) $-1/4$ to 1
- (c) 0 to 1
- (d) $-1/5$ to 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $-1/4$ to 1

EXAM SHORTCUT (AI-generated explanation)

The floor is $-1/(k - 1) = -1/4$.

16. FORECAST · FS-137 Correlation Ratio**QUESTION**

Which relation always holds between the correlation ratio η and the correlation coefficient r ?

OPTIONS

- (a) $\eta^2 \leq r^2$
- (b) $\eta^2 \geq r^2$
- (c) $\eta = r$
- (d) $\eta^2 = 1 - r^2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\eta^2 \geq r^2$

EXAM SHORTCUT (AI-generated explanation)

The correlation ratio measures all of the between-group variation, of which the linear part is r^2 .

17. **FORECAST · FS-138** Correlation Ratio**QUESTION**

The correlation ratio of Y on X equals r^2 if and only if

OPTIONS

- (a) $\rho = 0$
- (b) the regression of Y on X is linear
- (c) X and Y are independent
- (d) $\sigma_x = \sigma_y$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the regression of Y on X is linear

EXAM SHORTCUT (AI-generated explanation)

Equality means the best predictor is already linear, so nothing is lost by restricting to straight lines.

18. **FORECAST · FS-139** Correlation Ratio**QUESTION**

The correlation ratios of Y on X and of X on Y are

OPTIONS

- (a) always equal
- (b) generally unequal
- (c) summing to 1
- (d) multiplying to r^2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) generally unequal

EXAM SHORTCUT (AI-generated explanation)

The correlation ratio is asymmetric because it conditions on one variable only.

19. **FORECAST · FS-140** Rank Correlation**QUESTION**

With $n = 10$ and the sum of squared rank differences equal to 30, what is Spearman's rank correlation?

OPTIONS

- (a) 9/11
- (b) 0.7
- (c) 0.9
- (d) 10/11

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 9/11

EXAM SHORTCUT (AI-generated explanation)

$$1 - 6(30)/(10 \times 99) = 1 - 180/990 = 9/11.$$

20. **FORECAST · FS-141** Rank Correlation**QUESTION**

For $n = 2$, Spearman's rank correlation can only take the value

OPTIONS

- (a) 0
- (b) plus or minus 1
- (c) any value
- (d) 0.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) plus or minus 1

EXAM SHORTCUT (AI-generated explanation)

With two ranks the two orders are either identical or reversed, giving +1 or -1.

21. **FORECAST · FS-142** Rank Correlation**QUESTION**

For $n = 3$ with no ties, a Spearman rank correlation of exactly 0 is

OPTIONS

- (a) possible
- (b) impossible
- (c) possible only with ties
- (d) always the case

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) impossible

EXAM SHORTCUT (AI-generated explanation)

The attainable values are $1, 1/2, -1/2$ and -1 ; zero is not among them.

22. **FORECAST · FS-143** Rank Correlation**QUESTION**

The ranks $(1, 2, 3)$ are paired with $(1, 3, 2)$. What is Kendall's τ ?

OPTIONS

- (a) $1/3$
- (b) $2/3$
- (c) 0
- (d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/3$

EXAM SHORTCUT (AI-generated explanation)

Two concordant pairs and one discordant pair out of three give $(2 - 1)/3 = 1/3$.

23. **FORECAST · FS-144** Sampling Distribution of r **QUESTION**

Under the null hypothesis that ρ is zero, r^2 follows

OPTIONS

- (a) $\text{Beta}(1/2, (n-2)/2)$
- (b) $\text{Beta}(1, n-2)$
- (c) χ^2 with 1 degree of freedom
- (d) $U(0, 1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\text{Beta}(1/2, (n-2)/2)$

EXAM SHORTCUT (AI-generated explanation)

The exact null distribution of r^2 is Beta with parameters $1/2$ and $(n-2)/2$.

24. **FORECAST · FS-145** Sampling Distribution of r **QUESTION**

Under the null hypothesis that ρ is zero, what is $E(r^2)$?

OPTIONS

- (a) $1/n$
- (b) $1/(n-1)$
- (c) $1/(n-2)$
- (d) 0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/(n-1)$

EXAM SHORTCUT (AI-generated explanation)

The $\text{Beta}(1/2, (n-2)/2)$ mean is $(1/2)/((n-1)/2) = 1/(n-1)$.

25. FORECAST · FS-146 Sampling Distribution of r **QUESTION**

Under the null hypothesis that ρ is zero, what is the expectation of the reciprocal of $(1 - r^2)$?

OPTIONS

- (a) $(n - 2)/(n - 3)$
- (b) $(n - 3)/(n - 4)$
- (c) $(n - 1)/(n - 2)$
- (d) 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $(n - 3)/(n - 4)$

EXAM SHORTCUT (AI-generated explanation)

With $1 - r^2$ following $\text{Beta}((n - 2)/2, 1/2)$, the inverse moment is $(n - 3)/(n - 4)$.

26. FORECAST · FS-147 Sampling Distribution of r **QUESTION**

Under the null hypothesis that ρ is zero, the density of r is proportional to

OPTIONS

- (a) $(1 - r^2)^{(n-4)/2}$
- (b) $(1 - r^2)^{n-2}$
- (c) $(1 - r^2)^{1/2}$
- (d) e^{-r^2}

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $(1 - r^2)^{(n-4)/2}$

EXAM SHORTCUT (AI-generated explanation)

The exact null density of r is proportional to $(1 - r^2)^{(n-4)/2}$ on $(-1, 1)$.

27. FORECAST · FS-148 Standard Error of r **QUESTION**

With $r = 0.6$ and $n = 64$, what is the large-sample standard error of r ?

OPTIONS

- (a) 0.08
- (b) 0.1
- (c) 0.075
- (d) 0.125

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.08

EXAM SHORTCUT (AI-generated explanation)

$$SE(r) = (1 - r^2)/\sqrt{n} = 0.64/8 = 0.08.$$

28. FORECAST · FS-149 Standard Error of r **QUESTION**

With $r = 0.6$ and $n = 64$, what is the probable error of r ?

OPTIONS

- (a) 0.054
- (b) 0.08
- (c) 0.027
- (d) 0.1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.054

EXAM SHORTCUT (AI-generated explanation)

$$PE = 0.6745 \text{ times } SE = 0.6745(0.08) = 0.054.$$

29. **FORECAST · FS-150** Standard Error of r **QUESTION**

By the classical rule of thumb, a correlation is regarded as significant if r exceeds

OPTIONS

- (a) its probable error
- (b) six times its probable error
- (c) three times its probable error
- (d) twice its probable error

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) six times its probable error

EXAM SHORTCUT (AI-generated explanation)

The traditional criterion is r greater than 6 PE.

30. **FORECAST · FS-151** Computing r **QUESTION**

With $\text{Cov}(X, Y) = 12$, $\text{Var}(X) = 16$ and $\text{Var}(Y) = 25$, what is r ?

OPTIONS

- (a) 0.6
- (b) 0.48
- (c) 0.75
- (d) 0.03

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.6

EXAM SHORTCUT (AI-generated explanation)

$$r = 12 / \sqrt{16 \times 25} = 12 / 20 = 0.6.$$

31. **FORECAST · FS-152** Invariance of Correlation**QUESTION**

X is $U(-1, 1)$. What is $\text{Corr}(X, X^2)$?

OPTIONS

- (a) 1
- (b) 0
- (c) $1/2$
- (d) $1/\sqrt{3}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0

EXAM SHORTCUT (AI-generated explanation)

$\text{Cov}(X, X^2) = E(X^3) = 0$ by symmetry.

32. **FORECAST · FS-153** Computing r **QUESTION**

X , Y and Z are independent and identically distributed. What is $\text{Corr}(X + Y, Y + Z)$?

OPTIONS

- (a) $1/2$
- (b) $1/\sqrt{2}$
- (c) $1/3$
- (d) 0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/2$

EXAM SHORTCUT (AI-generated explanation)

The only shared variance is that of Y , giving $\sigma^2/(2\sigma^2) = 1/2$.

33. **FORECAST · FS-154** Invariance of Correlation**QUESTION**

What is $\text{Corr}(aX + b, cX + d)$ for non-zero a and c ?

OPTIONS

- (a) 1
- (b) the sign of ac
- (c) ac
- (d) 0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the sign of ac

EXAM SHORTCUT (AI-generated explanation)

Both are exact linear functions of the same X , so the correlation is plus or minus 1 according to the sign of ac .

34. **FORECAST · FS-155** Admissible Correlation Matrices**QUESTION**

With $r_{12} = 0.6$ and $r_{13} = 0.8$, what is the admissible range of r_{23} ?

OPTIONS

- (a) -1 to 1
- (b) 0 to 0.96
- (c) 0.48 to 1
- (d) -0.48 to 0.48

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0 to 0.96

EXAM SHORTCUT (AI-generated explanation)

The bound is $r_{12}r_{13}$ plus or minus $\sqrt{(1 - r_{12}^2)(1 - r_{13}^2)} = 0.48$ plus or minus 0.48.

35. FORECAST · FS-156 Admissible Correlation Matrices**QUESTION**

Three variables have a common pairwise correlation r . This is admissible if and only if

OPTIONS

- (a) $r \geq 0$
- (b) $r \geq -1/2$
- (c) $|r| \leq 1/2$
- (d) $r \geq -1/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $r \geq -1/2$

EXAM SHORTCUT (AI-generated explanation)

The variance of the sum is $3(1 + 2r)\sigma^2$, which must be non-negative.

36. FORECAST · FS-157 Partial Correlation**QUESTION**

With $r_{12} = r_{13} = r_{23} = 0.5$, what is $r_{12.3}$?

OPTIONS

- (a) $1/3$
- (b) $1/2$
- (c) 0
- (d) $1/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/3$

EXAM SHORTCUT (AI-generated explanation)

The equicorrelation formula $r/(1 + r)$ gives $0.5/1.5 = 1/3$.

37. FORECAST · FS-158 Multiple Correlation**QUESTION**

Which identity holds for the multiple correlation of variable 1 on variables 2 and 3?

OPTIONS

- (a) $1 - R^2$ equals $(1 - r_{12}^2)(1 - r_{13}^2)$
- (b) R^2 equals the product of r_{12}^2 and r_{13}^2
- (c) $1 - R^2$ equals $1 - r_{12}^2 - r_{13}^2$
- (d) none of these

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1 - R^2$ equals $(1 - r_{12}^2)(1 - r_{13}^2)$

EXAM SHORTCUT (AI-generated explanation)

The unexplained variation multiplies: first remove variable 2, then remove variable 3 from what is left.

38. FORECAST · FS-159 Partial Correlation**QUESTION**

If $r_{12.3} = 0$, then

OPTIONS

- (a) $r_{12} = 0$
- (b) $r_{12} = r_{13}r_{23}$
- (c) $r_{13} = 0$
- (d) $r_{23} = 0$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $r_{12} = r_{13}r_{23}$

EXAM SHORTCUT (AI-generated explanation)

The numerator of the ∂ correlation vanishes exactly when r_{12} equals the product of the other two.

39. **FORECAST · FS-160** Rank Correlation**QUESTION**

For 10 pairs of observations there are 9 pairs of consecutive deviations, of which 7 are concurrent. What is the coefficient of concurrent deviations?

OPTIONS

- (a) 0.745
- (b) 0.4
- (c) 0.55
- (d) 0.67

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.745

EXAM SHORTCUT (AI-generated explanation)

The formula gives the square root of $(2c - m)/m$ with $m = 9$ and $c = 7$: $\sqrt{5/9} = 0.745$.

40. **FORECAST · FS-161** Invariance of Correlation**QUESTION**

Two variables have $r = 0$. Which conclusion is warranted?

OPTIONS

- (a) they are independent
- (b) there is no linear association between them
- (c) there is no association of any kind between them
- (d) one is a constant

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) there is no linear association between them

EXAM SHORTCUT (AI-generated explanation)

A zero correlation rules out linear association only; any curved relation may still be present.

41. **FORECAST · FS-162** Fisher Z Transformation**QUESTION**

Testing the hypothesis that ρ equals 0.5 with $r = 0.7$ and $n = 28$ gives a z statistic of about

OPTIONS

- (a) 1.59
- (b) 2.5
- (c) 0.8
- (d) 3.1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 1.59

EXAM SHORTCUT (AI-generated explanation)

$Z(0.7) = 0.8673$ and $Z(0.5) = 0.5493$, and dividing the difference by 0.2 gives 1.59.

42. **FORECAST · FS-163** Fisher Z Transformation**QUESTION**

For two independent samples each of size 28, what is the standard error of the difference of the two Fisher Z values?

OPTIONS

- (a) 0.2
- (b) 0.283
- (c) 0.4
- (d) 0.1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0.283

EXAM SHORTCUT (AI-generated explanation)

$\sqrt{1/25 + 1/25} = \sqrt{0.08} = 0.283.$

43. **FORECAST · FS-164** Computing r **QUESTION**

The coefficient of alienation equals

OPTIONS

- (a) $1 - r^2$
- (b) the square root of $(1 - r^2)$
- (c) $1 - r$
- (d) $1/r$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the square root of $(1 - r^2)$

EXAM SHORTCUT (AI-generated explanation)

The coefficient of alienation is $\sqrt{1 - r^2}$, the standardised residual standard deviation.

44. **FORECAST · FS-165** Computing r **QUESTION**

With $b_{yx} = 0.5$ and $\sigma_y = 2\sigma_x$, what is r ?

OPTIONS

- (a) 1
- (b) 0.25
- (c) 0.5
- (d) 0.125

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0.25

EXAM SHORTCUT (AI-generated explanation)

$r = b_{yx}\sigma_x/\sigma_y = 0.5(1/2) = 0.25$.

45. **FORECAST · FS-166** Correlation Ratio**QUESTION**

The squared correlation ratio of Y on X equals

OPTIONS

- (a) the variance of $E(Y \text{ given } X)$ divided by $\text{Var}(Y)$
- (b) the expectation of $\text{Var}(Y \text{ given } X)$ divided by $\text{Var}(Y)$
- (c) the squared covariance divided by the product of the variances
- (d) $\text{Var}(Y)$ divided by $\text{Var}(X)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) the variance of $E(Y \text{ given } X)$ divided by $\text{Var}(Y)$

EXAM SHORTCUT (AI-generated explanation)

It is the between-group share of the total variance.

46. **FORECAST · FS-167** Correlation Ratio**QUESTION**

The squared correlation ratio of Y on X equals 1 if and only if

OPTIONS

- (a) Y is an exact function of X
- (b) X and Y are independent
- (c) $r = 1$
- (d) X is an exact function of Y

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) Y is an exact function of X

EXAM SHORTCUT (AI-generated explanation)

A unit correlation ratio means the within-group variance vanishes, so Y is determined by X.

47. FORECAST · FS-168 Intraclass Correlation**QUESTION**

The intraclass correlation attains its minimum value when

OPTIONS

- (a) the group variance equals the total variance
- (b) the between-group variance is zero
- (c) the total variance is zero
- (d) k equals 2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the between-group variance is zero

EXAM SHORTCUT (AI-generated explanation)

The coefficient decreases as the between-group variance falls, reaching $-1/(k-1)$ at zero.

48. FORECAST · FS-169 Multiple Correlation**QUESTION**

With all three pairwise correlations equal to 0.7, what is the squared multiple correlation of variable 1 on 2 and 3?

OPTIONS

- (a) 0.576
- (b) 0.49
- (c) 0.7
- (d) 0.98

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.576

EXAM SHORTCUT (AI-generated explanation)

$2r^2/(1+r) = 2(0.49)/1.7 = 0.576.$

49. **FORECAST · FS-170** Multiple Correlation**QUESTION**

If $r_{23} = 0$, the squared multiple correlation of variable 1 on 2 and 3 equals

OPTIONS

- (a) $r_{12}^2 + r_{13}^2$
- (b) r_{12}^2
- (c) 0
- (d) $r_{12}r_{13}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $r_{12}^2 + r_{13}^2$

EXAM SHORTCUT (AI-generated explanation)

Orthogonal predictors contribute independently, so the explained fractions simply add.

50. **FORECAST · FS-171** Partial Correlation**QUESTION**

If $r_{13} = r_{23} = 0$, then $r_{12.3}$ equals

OPTIONS

- (a) 0
- (b) r_{12}
- (c) 1
- (d) undefined

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) r_{12}

EXAM SHORTCUT (AI-generated explanation)

With variable 3 unrelated to both others, controlling for it changes nothing.

51. **FORECAST · FS-172** Computing r **QUESTION**

The pairs $(1, 2)$, $(2, 4)$ and $(3, 6)$ are observed. What is r ?

OPTIONS

- (a) 1
- (b) 0.5
- (c) 2
- (d) 0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 1

EXAM SHORTCUT (AI-generated explanation)

The points lie exactly on $y = 2x$, so the correlation is 1.

52. **FORECAST · FS-173** Computing r **QUESTION**

The pairs $(-1, 1)$, $(0, 0)$ and $(1, 1)$ are observed. What is r ?

OPTIONS

- (a) 1
- (b) 0
- (c) -1
- (d) 0.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0

EXAM SHORTCUT (AI-generated explanation)

The configuration is symmetric about $x = 0$, so the covariance vanishes.

53. **FORECAST · FS-174** Fisher Z Transformation**QUESTION**

With $r = 0.6$ and $n = 28$, an approximate 95% confidence interval for ρ is

OPTIONS

- (a) 0.29 to 0.80
- (b) 0.4 to 0.8
- (c) 0.5 to 0.7
- (d) 0.2 to 1.0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.29 to 0.80

EXAM SHORTCUT (AI-generated explanation)

Work on the Z scale: 0.6931 plus or minus 1.96(0.2) gives 0.301 to 1.085, which maps back to 0.29 to 0.80.

54. **FORECAST · FS-175** Fisher Z Transformation**QUESTION**

Fisher's Z transformation is preferred because

OPTIONS

- (a) it is exact for small samples
- (b) its variance is almost free of ρ
- (c) it is unbiased
- (d) it needs no normality assumption

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) its variance is almost free of ρ

EXAM SHORTCUT (AI-generated explanation)

The variance $1/(n - 3)$ does not involve ρ , which makes tests and intervals straightforward.

55. **FORECAST · FS-176** Partial Correlation**QUESTION**

Comparing the ∂ correlation $r_{12.3}$ with the simple correlation r_{12} ,

OPTIONS

- (a) the ∂ is always at most the simple in absolute value
- (b) the ∂ can exceed the simple
- (c) they are always equal
- (d) they always have opposite signs

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the ∂ can exceed the simple

EXAM SHORTCUT (AI-generated explanation)

Controlling for a suppressor variable can raise the association considerably.

56. **FORECAST · FS-177** Multiple Correlation**QUESTION**

In a standardised regression of Y on X_2 and X_3 with β weights β_2 and β_3 , the squared multiple correlation equals

OPTIONS

- (a) $\beta_2 r_{12} + \beta_3 r_{13}$
- (b) $\beta_2^2 + \beta_3^2$
- (c) $\beta_2 \beta_3$
- (d) $r_{12} + r_{13}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\beta_2 r_{12} + \beta_3 r_{13}$

EXAM SHORTCUT (AI-generated explanation)

R^2 is the sum of the β weights times the corresponding simple correlations.

57. **FORECAST · FS-178** Rank Correlation**QUESTION**

With $n = 4$ and the sum of squared rank differences equal to 20, what is Spearman's rank correlation?

OPTIONS

- (a) 0
- (b) -1
- (c) 1
- (d) 0.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) -1

EXAM SHORTCUT (AI-generated explanation)

$1 - 6(20)/(4 \times 15) = 1 - 2 = -1$, the maximum possible sum of squared differences.

58. **FORECAST · FS-179** Rank Correlation**QUESTION**

Kendall's τ and Spearman's ρ are

OPTIONS

- (a) always equal
- (b) generally different, both lying between -1 and 1
- (c) such that τ is at least ρ
- (d) related by ρ equal to τ squared

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) generally different, both lying between -1 and 1

EXAM SHORTCUT (AI-generated explanation)

They measure concordance differently and generally take different values on the same data.

59. **FORECAST · FS-180** Computing r **QUESTION**

Consider: 1. r is free of units, 2. r measures any form of dependence, 3. $|r| = 1$ if and only if there is an exact linear relation. Which are correct?

OPTIONS

- (a) 1 and 3
- (b) 1 and 2
- (c) all three
- (d) 2 only

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 1 and 3

EXAM SHORTCUT (AI-generated explanation)

r is unit-free and characterises exact linear relations, but it captures only the linear part of any dependence.

60. **FORECAST · FS-181** Standard Error of r **QUESTION**

For large n , the variance of the sample correlation is approximately

OPTIONS

- (a) $(1 - \rho^2)^2/n$
- (b) $(1 - \rho^2)/n$
- (c) $1/n$
- (d) ρ^2/n

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $(1 - \rho^2)^2/n$

EXAM SHORTCUT (AI-generated explanation)

The asymptotic variance of r is $(1 - \rho^2)^2/n$.

61. **FORECAST · FS-330** Multiple Correlation**QUESTION**

With the squared multiple correlation of variable 1 on 2 and 3 equal to 0.64 and $r_{12} = 0.6$, what is the absolute value of r_{13} ?

OPTIONS

- (a) 0.661
- (b) 0.8
- (c) 0.5
- (d) 0.36

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.661

EXAM SHORTCUT (AI-generated explanation)

$1 - 0.64 = (1 - 0.36)(1 - x^2)$ gives $x^2 = 0.4375$, so $x = 0.661$.

62. **FORECAST · FS-331** Sampling Distribution of r **QUESTION**

Under the null hypothesis that ρ is zero, what is the exact variance of r ?

OPTIONS

- (a) $1/(n - 1)$
- (b) $1/n$
- (c) $(1 - r^2)/n$
- (d) $1/(n - 3)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/(n - 1)$

EXAM SHORTCUT (AI-generated explanation)

$E(r) = 0$ and $E(r^2) = 1/(n - 1)$, so the variance is $1/(n - 1)$.

63. **FORECAST · FS-344** Testing a Correlation**QUESTION**

With $n = 12$ and $r = 0.5$, and a critical t value of 2.228 on 10 degrees of freedom, is the correlation significant at 5%?

OPTIONS

- (a) yes
- (b) no, because t is about 1.83
- (c) yes, because t exceeds 2.5
- (d) the test cannot be applied

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) no, because t is about 1.83

EXAM SHORTCUT (AI-generated explanation)

$$t = 0.5\sqrt{10}/\sqrt{0.75} = 1.826, \text{ which is below } 2.228.$$

64. **FORECAST · FS-345** Multiple Correlation**QUESTION**

With $r_{12} = 0.6$, $r_{13} = 0.8$ and $r_{23} = 0$, what is the multiple correlation of variable 1 on variables 2 and 3?

OPTIONS

- (a) 1
- (b) 0.8
- (c) 0.64
- (d) 0.98

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 1

EXAM SHORTCUT (AI-generated explanation)

With orthogonal predictors $R^2 = 0.36 + 0.64 = 1$, so variable 1 is an exact linear function of the other two.

65. **FORECAST · FS-350** Intraclass Correlation**QUESTION**

With $k = 3$ members per family and all family means equal, the intraclass correlation is

OPTIONS

- (a) $-1/2$
- (b) 0
- (c) 1
- (d) $-1/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $-1/2$

EXAM SHORTCUT (AI-generated explanation)

Zero between-family variance drives the coefficient to its floor $-1/(k - 1) = -1/2$.

66. **FORECAST · FS-351** Rank Correlation**QUESTION**

With $n = 5$ and the sum of squared rank differences equal to 40, what is Spearman's rank correlation?

OPTIONS

- (a) -1
- (b) 0
- (c) 1
- (d) -0.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) -1

EXAM SHORTCUT (AI-generated explanation)

The maximum sum of d^2 is $n(n^2 - 1)/3 = 40$, attained only by a complete reversal.

67. **FORECAST · FS-352** Rank Correlation**QUESTION**

The ranks (1, 2, 3, 4) are paired with (2, 1, 4, 3). What is Kendall's τ ?

OPTIONS

- (a) $1/3$
- (b) $1/2$
- (c) $2/3$
- (d) 0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/3$

EXAM SHORTCUT (AI-generated explanation)

Four concordant and two discordant pairs out of six give $(4 - 2)/6 = 1/3$.

68. **FORECAST · FS-363** Intraclass Correlation**QUESTION**

For an intraclass correlation with $k = 2$, the admissible range is

OPTIONS

- (a) -1 to 1
- (b) $-1/2$ to 1
- (c) 0 to 1
- (d) $-1/3$ to 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) -1 to 1

EXAM SHORTCUT (AI-generated explanation)

The floor $-1/(k - 1)$ equals -1 when $k = 2$.

69. **FORECAST · FS-371** Computing r

QUESTION

Two variables are strongly correlated because both increase with time. This illustrates

OPTIONS

- (a) causation
- (b) spurious correlation induced by a common cause
- (c) zero correlation
- (d) a Type I error

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) spurious correlation induced by a common cause

EXAM SHORTCUT (AI-generated explanation)

A shared trend produces correlation without any direct link between the two series.

FORECAST · STATISTICAL METHODS

Curve Fitting and Orthogonal Polynomials

7 forecast problems · 2 concepts

1. **FORECAST · FS-057** Orthogonal Polynomials

QUESTION

The fourth-degree Hermite polynomial associated with the standard normal distribution is

OPTIONS

- (a) $x^4 - 6x^2 + 3$
- (b) $x^4 - 3x^2$
- (c) $x^4 - 6x^2$
- (d) $x^4 - 4x^2 + 2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $x^4 - 6x^2 + 3$

EXAM SHORTCUT (AI-generated explanation)

The probabilists' Hermite polynomials are $1, x, x^2 - 1, x^3 - 3x, x^4 - 6x^2 + 3$.

2. **FORECAST · FS-058** Curve Fitting

QUESTION

To fit $y = ab^x$ by least squares, one linearises by regressing

OPTIONS

- (a) $\log y$ on x
- (b) $\log y$ on $\log x$
- (c) y on $1/x$
- (d) $1/y$ on x

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\log y$ on x

EXAM SHORTCUT (AI-generated explanation)

Taking logarithms gives $\log y = \log a + x \log b$, which is linear in x .

3. FORECAST · FS-059 Curve Fitting**QUESTION**

To fit $y = ax^b$ by least squares, one linearises by regressing

OPTIONS

- (a) $\log y$ on x
- (b) $\log y$ on $\log x$
- (c) y on x^2
- (d) $\sqrt{y}$ on x

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\log y$ on $\log x$

EXAM SHORTCUT (AI-generated explanation)

Taking logarithms gives $\log y = \log a + b \log x$, which is linear in $\log x$.

4. FORECAST · FS-060 Curve Fitting**QUESTION**

Model E_1 has residual sum of squares 12 and model E_2 has 8, with the same number of parameters. For prediction one prefers

OPTIONS

- (a) E_1
- (b) E_2
- (c) either
- (d) it cannot be decided

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) E_2

EXAM SHORTCUT (AI-generated explanation)

With the same complexity, the smaller residual sum of squares means the better fit.

5. FORECAST · FS-061 Curve Fitting**QUESTION**

Fitting a polynomial of degree k by least squares requires how many normal equations?

OPTIONS

- (a) k
- (b) $k + 1$
- (c) $2k$
- (d) $k - 1$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $k + 1$

EXAM SHORTCUT (AI-generated explanation)

A degree- k polynomial has $k + 1$ coefficients, hence $k + 1$ normal equations.

6. FORECAST · FS-062 Orthogonal Polynomials**QUESTION**

The advantage of using orthogonal polynomials in curve fitting is that

OPTIONS

- (a) fewer data points are needed
- (b) adding a higher-degree term leaves the earlier coefficients unchanged
- (c) no normal equations arise
- (d) the residuals become zero

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) adding a higher-degree term leaves the earlier coefficients unchanged

EXAM SHORTCUT (AI-generated explanation)

Orthogonality decouples the normal equations, so each coefficient is computed independently of the others.

7. FORECAST · FS-063 Curve Fitting**QUESTION**

A least-squares line fitted with an intercept always satisfies

OPTIONS

- (a) the sum of the residuals is zero
- (b) the sum of x times the residuals is zero
- (c) both of these
- (d) the sum of the squared residuals is zero

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) both of these

EXAM SHORTCUT (AI-generated explanation)

The two normal equations are exactly these two orthogonality conditions.

FORECAST · STATISTICAL METHODS

Data Presentation and Measurement Scales

11 forecast problems · 2 concepts

1. FORECAST · FS-001 Scales of Measurement**QUESTION**

Temperature measured in degrees Celsius is on which scale?

OPTIONS

- (a) nominal
- (b) ordinal
- (c) interval
- (d) ratio

ANSWER (AI-generated forecast item — no official UPSC key exists)**(c)** interval**EXAM SHORTCUT** (AI-generated explanation)

Differences are meaningful but the zero is arbitrary, so the scale is interval, not ratio.

2. FORECAST · FS-002 Scales of Measurement**QUESTION**

For nominal data, the only permissible average is the

OPTIONS

- (a) arithmetic mean
- (b) median
- (c) mode
- (d) geometric mean

ANSWER (AI-generated forecast item — no official UPSC key exists)**(c)** mode**EXAM SHORTCUT** (AI-generated explanation)

Nominal categories have no order, so only the most frequent category is meaningful.

3. FORECAST · FS-003 Frequency Distributions**QUESTION**

Which of the following condense raw data: 1. an array, 2. a frequency distribution, 3. a histogram, 4. an ogive?

OPTIONS

- (a) 2 only
- (b) 2 and 3
- (c) 2, 3 and 4
- (d) all four

ANSWER (AI-generated forecast item — no official UPSC key exists)

(d) all four

EXAM SHORTCUT (AI-generated explanation)

Every one of them replaces the raw list by a more compact summary, an ordered array included.

4. FORECAST · FS-004 Frequency Distributions**QUESTION**

The median and the quartiles are read graphically from

OPTIONS

- (a) a histogram
- (b) a frequency polygon
- (c) an ogive
- (d) $\bar{a}$ chart

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) an ogive

EXAM SHORTCUT (AI-generated explanation)

The ogive is the cumulative curve, so any quantile is read off at the corresponding cumulative frequency.

5. FORECAST · FS-005 Frequency Distributions**QUESTION**

In a histogram with unequal class widths the heights of the bars must be proportional to

OPTIONS

- (a) the frequency
- (b) the frequency density
- (c) the cumulative frequency
- (d) the relative frequency

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the frequency density

EXAM SHORTCUT (AI-generated explanation)

Area must represent frequency, so the height is the frequency divided by the class width.

6. FORECAST · FS-006 Scales of Measurement**QUESTION**

The Kelvin temperature scale is

OPTIONS

- (a) interval
- (b) ratio
- (c) ordinal
- (d) nominal

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) ratio

EXAM SHORTCUT (AI-generated explanation)

Kelvin has an absolute zero, so ratios are meaningful and the scale is ratio.

7. FORECAST · FS-007 Frequency Distributions**QUESTION**

By Sturges' rule, the number of classes for $n = 1000$ observations is approximately

OPTIONS

- (a) 8
- (b) 10
- (c) 11
- (d) 13

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 11

EXAM SHORTCUT (AI-generated explanation)

$1 + 3.322 \log_{10}(1000) = 1 + 9.97 = 10.97$, so about 11 classes.

8. FORECAST · FS-008 Scales of Measurement**QUESTION**

Consider: 1. Likert scores are ordinal, 2. postal index numbers are on a ratio scale, 3. income is on a ratio scale. Which are correct?

OPTIONS

- (a) 1 only
- (b) 1 and 3
- (c) 2 and 3
- (d) all three

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 1 and 3

EXAM SHORTCUT (AI-generated explanation)

Likert scores are ordered categories and income has a true zero; postal codes are mere labels and are nominal.

9. FORECAST · FS-009 Frequency Distributions**QUESTION**

In a pie chart with total frequency 360, a class of frequency 45 occupies a sector of

OPTIONS

- (a) 12.5 degrees
- (b) 45 degrees
- (c) 90 degrees
- (d) 30 degrees

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 45 degrees

EXAM SHORTCUT (AI-generated explanation)

The angle is $360 \times \frac{45}{360} = 45$ degrees.

10. FORECAST · FS-010 Frequency Distributions**QUESTION**

A frequency polygon is obtained by joining

OPTIONS

- (a) the lower class limits
- (b) the midpoints of the tops of the histogram bars
- (c) the cumulative frequencies
- (d) the upper class limits

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the midpoints of the tops of the histogram bars

EXAM SHORTCUT (AI-generated explanation)

The polygon connects the class midpoints at the corresponding frequencies.

11. **FORECAST · FS-375** Frequency Distributions**QUESTION**

In a symmetric unimodal distribution, the mean, the median and the mode

OPTIONS

- (a) all coincide
- (b) satisfy $\text{mean} > \text{median} > \text{mode}$
- (c) satisfy $\text{mode} > \text{median} > \text{mean}$
- (d) are unrelated

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) all coincide

EXAM SHORTCUT (AI-generated explanation)

Symmetry puts the mean and median at the centre of symmetry, and unimodality puts the mode there too.

FORECAST · STATISTICAL METHODS

Measures of Location, Dispersion, Skewness and Kurtosis

35 forecast problems · 6 concepts

1. FORECAST · FS-011 Skewness**QUESTION**

The mean is 50, the median 48 and the standard deviation 6. What is Pearson's coefficient of skewness in its median form?

OPTIONS

- (a) 0.33
- (b) 1
- (c) 0.5
- (d) 2

ANSWER (AI-generated forecast item — no official UPSC key exists)**(b) 1****EXAM SHORTCUT** (AI-generated explanation)

$$3(50 - 48)/6 = 1.$$

2. FORECAST · FS-012 Skewness**QUESTION**

The mode is 40, the mean 46 and Pearson's coefficient of skewness is 0.5. What is the standard deviation?

OPTIONS

- (a) 6
- (b) 12
- (c) 3
- (d) 24

ANSWER (AI-generated forecast item — no official UPSC key exists)**(b) 12****EXAM SHORTCUT** (AI-generated explanation)

The mode form gives $(46 - 40)/SD = 0.5$, so $SD = 12$.

3. FORECAST · FS-013 Combined Mean and Variance**QUESTION**

Group 1 has 40 observations with mean 60 and group 2 has 60 observations with mean 70. What is the combined mean?

OPTIONS

- (a) 65
- (b) 66
- (c) 64
- (d) 67

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 66

EXAM SHORTCUT (AI-generated explanation)

$$(40 \times 60 + 60 \times 70)/100 = 6600/100 = 66.$$

4. FORECAST · FS-014 Combined Mean and Variance**QUESTION**

Two groups of 50 observations each have means 10 and 20 and standard deviations 3 and 4. What is the combined variance?

OPTIONS

- (a) 12.5
- (b) 25
- (c) 37.5
- (d) 50

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 37.5

EXAM SHORTCUT (AI-generated explanation)

Within part $(9 + 16)/2 = 12.5$ plus between part 25 gives 37.5.

5. FORECAST · FS-015 Combined Mean and Variance**QUESTION**

A combined sample of 100 has mean 50 and variance 100. The first group has 60 observations with mean 45 and variance 40. What is the variance of the second group?

OPTIONS

- (a) 96.25
- (b) 60
- (c) 190
- (d) 100

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 96.25

EXAM SHORTCUT (AI-generated explanation)

The second mean is 57.5; solving $10000 = 60(40 + 25) + 40(v + 56.25)$ gives $v = 96.25$.

6. FORECAST · FS-016 Measures of Dispersion**QUESTION**

If $\text{Var}(X) = 4$, what is $\text{Var}(3X - 5)$?

OPTIONS

- (a) 7
- (b) 12
- (c) 36
- (d) 31

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 36

EXAM SHORTCUT (AI-generated explanation)

Variance scales with the square of the multiplier and ignores the shift: $9(4) = 36$.

7. FORECAST · FS-017 Measures of Dispersion**QUESTION**

Height has mean 160 and standard deviation 8; weight has mean 60 and standard deviation 6. Weight is how many times as variable as height?

OPTIONS

- (a) 0.75
- (b) 1.33
- (c) 2
- (d) 0.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 2

EXAM SHORTCUT (AI-generated explanation)

Coefficients of variation are 5% and 10%, so weight is twice as variable.

8. FORECAST · FS-018 Measures of Dispersion**QUESTION**

The mean deviation is a minimum when taken about the

OPTIONS

- (a) mean
- (b) median
- (c) mode
- (d) origin

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) median

EXAM SHORTCUT (AI-generated explanation)

The sum of absolute deviations is minimised at the median.

9. FORECAST · FS-019 Measures of Dispersion**QUESTION**

The root-mean-square deviation is a minimum when taken about the

OPTIONS

- (a) mean
- (b) median
- (c) mode
- (d) upper quartile

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) mean

EXAM SHORTCUT (AI-generated explanation)

The sum of squared deviations is minimised at the arithmetic mean.

10. FORECAST · FS-020 Measures of Dispersion**QUESTION**

For a normal distribution the ratio of quartile deviation to mean deviation to standard deviation is approximately

OPTIONS

- (a) 2 : 3 : 4
- (b) 10 : 12 : 15
- (c) 1 : 2 : 3
- (d) 3 : 4 : 5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 10 : 12 : 15

EXAM SHORTCUT (AI-generated explanation)

0.6745 : 0.7979 : 1 scales to about 10 : 12 : 15.

11. **FORECAST · FS-021** Kurtosis**QUESTION**

What is the value of β_2 for the uniform distribution on $(0, 1)$?

OPTIONS

- (a) 3
- (b) 1.8
- (c) 2.4
- (d) 1.2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 1.8

EXAM SHORTCUT (AI-generated explanation)

$\mu_4 = 1/80$ and $\mu_2 = 1/12$, so $\beta_2 = (1/80)(144) = 1.8$.

12. **FORECAST · FS-022** Kurtosis**QUESTION**

Which relation always holds between β_1 and β_2 ?

OPTIONS

- (a) $\beta_2 \geq \beta_1$
- (b) $\beta_2 \geq \beta_1 + 1$
- (c) $\beta_2 \leq \beta_1 + 1$
- (d) $\beta_2 = \beta_1 + 3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\beta_2 \geq \beta_1 + 1$

EXAM SHORTCUT (AI-generated explanation)

The Cauchy-Schwarz inequality applied to the second and third central moments gives $\beta_2 \geq \beta_1 + 1$.

13. FORECAST · FS-023 Skewness**QUESTION**

If $\text{mean} > \text{median} > \text{mode}$, the distribution is

OPTIONS

- (a) symmetric
- (b) negatively skewed
- (c) positively skewed
- (d) mesokurtic

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) positively skewed

EXAM SHORTCUT (AI-generated explanation)

The mean is dragged furthest by the long right tail, so the ordering signals positive skewness.

14. FORECAST · FS-024 Skewness**QUESTION**

With $Q_1 = 10$, $Q_3 = 30$ and median 18, what is Bowley's coefficient of skewness?

OPTIONS

- (a) 0.2
- (b) 0.4
- (c) -0.2
- (d) 0.1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.2

EXAM SHORTCUT (AI-generated explanation)

$(30 + 10 - 36)/(30 - 10) = 4/20 = 0.2.$

15. **FORECAST · FS-025** Skewness**QUESTION**

Kelly's coefficient of skewness is based on

OPTIONS

- (a) the quartiles
- (b) the 10th and 90th percentiles
- (c) the 2nd and 8th deciles
- (d) the 5th and 95th percentiles

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the 10th and 90th percentiles

EXAM SHORTCUT (AI-generated explanation)

Kelly's measure uses P_{10} , P_{50} and P_{90} , widening Bowley's quartile-based measure.

16. **FORECAST · FS-026** Measures of Location**QUESTION**

What is the harmonic mean of 2, 4 and 8?

OPTIONS

- (a) 4
- (b) $24/7$
- (c) $14/3$
- (d) 3

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $24/7$

EXAM SHORTCUT (AI-generated explanation)

$3/(1/2 + 1/4 + 1/8) = 3/(7/8) = 24/7.$

17. **FORECAST · FS-027** Measures of Location**QUESTION**

For two positive numbers, which identity holds?

OPTIONS

- (a) AM times HM equals GM squared
- (b) AM times GM equals HM squared
- (c) GM times HM equals AM squared
- (d) none of these

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) AM times HM equals GM squared

EXAM SHORTCUT (AI-generated explanation)

For two numbers a and b , $AM \times HM = [(a+b)/2][2ab/(a+b)] = ab$, which is the square of the GM.

18. **FORECAST · FS-028** Measures of Dispersion**QUESTION**

What is the variance of the first 10 natural numbers?

OPTIONS

- (a) 8.25
- (b) 9.17
- (c) 5.5
- (d) 33

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 8.25

EXAM SHORTCUT (AI-generated explanation)

$(N^2 - 1)/12 = 99/12 = 8.25$.

19. **FORECAST · FS-029** Measures of Dispersion**QUESTION**

Every observation is multiplied by -2. The standard deviation becomes

OPTIONS

- (a) -2σ
- (b) 2σ
- (c) 4σ
- (d) σ

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 2σ

EXAM SHORTCUT (AI-generated explanation)

The standard deviation uses $|a|$, so it becomes 2σ , not -2σ .

20. **FORECAST · FS-030** Measures of Dispersion**QUESTION**

Sheppard's correction to the second moment for a grouped distribution with class width h is

OPTIONS

- (a) add $h^2/12$
- (b) subtract $h^2/12$
- (c) subtract $h^2/24$
- (d) add $h/12$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) subtract $h^2/12$

EXAM SHORTCUT (AI-generated explanation)

Grouping inflates the variance, so the correction subtracts $h^2/12$.

21. **FORECAST · FS-031** Moments**QUESTION**

With the first three raw moments 2, 6 and 20, what is the third central moment?

OPTIONS

- (a) 0
- (b) 4
- (c) -4
- (d) 8

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0

EXAM SHORTCUT (AI-generated explanation)

$$20 - 3(2)(6) + 2(8) = 20 - 36 + 16 = 0.$$

22. **FORECAST · FS-032** Skewness**QUESTION**

With $\mu_2 = 4$ and $\mu_3 = 8$, what is β_1 ?

OPTIONS

- (a) 2
- (b) 1
- (c) 0.5
- (d) 4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 1

EXAM SHORTCUT (AI-generated explanation)

$$\beta_1 = \mu_3^2 / \mu_2^3 = 64 / 64 = 1.$$

23. **FORECAST · FS-033** Skewness**QUESTION**

Pearson's coefficient of skewness lies in the range

OPTIONS

- (a) -1 to 1
- (b) -3 to 3
- (c) 0 to 1
- (d) it is unbounded

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) -3 to 3

EXAM SHORTCUT (AI-generated explanation)

The mode form is bounded by 1 in absolute value, so the median form, being three times a quantity bounded by 1, lies in -3 to 3.

24. **FORECAST · FS-034** Measures of Location**QUESTION**

With $N = 100$, a median class of 20 to 30, a cumulative frequency of 35 below it and a class frequency of 30, what is the median?

OPTIONS

- (a) 25
- (b) 24
- (c) 26
- (d) 22

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 25

EXAM SHORTCUT (AI-generated explanation)

$20 + (50 - 35)(10)/30 = 20 + 5 = 25.$

25. **FORECAST · FS-035** Measures of Location**QUESTION**

With a modal class of 30 to 40, a preceding frequency of 10, a modal frequency of 20 and a following frequency of 14, what is the mode?

OPTIONS

- (a) 35
- (b) 36.25
- (c) 34
- (d) 37.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 36.25

EXAM SHORTCUT (AI-generated explanation)

$$30 + [(20 - 10)/((20 - 10) + (20 - 14))](10) = 30 + 100/16 = 36.25.$$

26. **FORECAST · FS-036** Measures of Dispersion**QUESTION**

What is the standard deviation of just two numbers a and b?

OPTIONS

- (a) $|a - b|$
- (b) $|a - b|/2$
- (c) $|a - b|/\sqrt{2}$
- (d) $(a + b)/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $|a - b|/2$

EXAM SHORTCUT (AI-generated explanation)

Each deviates from the mean by $|a - b|/2$, so the standard deviation is $|a - b|/2$.

27. FORECAST · FS-037 Kurtosis**QUESTION**

For a Poisson distribution with mean 4, what is β_2 ?

OPTIONS

- (a) 3
- (b) 3.25
- (c) 4
- (d) 7

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 3.25

EXAM SHORTCUT (AI-generated explanation)

$$\beta_2 = 3 + 1/\lambda = 3 + 0.25.$$

28. FORECAST · FS-038 Measures of Dispersion**QUESTION**

Consider: 1. the mean deviation never exceeds the standard deviation, 2. the sum of squared deviations about any constant A is at least the sum about the mean, 3. the range is unaffected by extreme values. Which are correct?

OPTIONS

- (a) 1 and 2
- (b) 1 only
- (c) 2 and 3
- (d) all three

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 1 and 2

EXAM SHORTCUT (AI-generated explanation)

Statements 1 and 2 are standard results; the range is determined entirely by the two extremes.

29. **FORECAST · FS-328** Combined Mean and Variance**QUESTION**

Three groups of 10 observations each have means 10, 20 and 30 and variance 4 apiece. What is the combined variance?

OPTIONS

- (a) $212/3$
- (b) 4
- (c) $200/3$
- (d) 100

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $212/3$

EXAM SHORTCUT (AI-generated explanation)

Within 4 plus between $(100 + 0 + 100)/3$ gives $4 + 200/3 = 212/3$.

30. **FORECAST · FS-348** Kurtosis**QUESTION**

With grouped moments $\mu_2 = 12$ and $\mu_4 = 400$ and class width 6, what is β_2 after Sheppard's corrections?

OPTIONS

- (a) 2.74
- (b) 2.78
- (c) 3.09
- (d) 2.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 2.74

EXAM SHORTCUT (AI-generated explanation)

Corrected $\mu_2 = 9$ and $\mu_4 = 400 - 216 + 37.8 = 221.8$, giving $221.8/81 = 2.74$.

31. **FORECAST · FS-349** Skewness**QUESTION**

The ranges of Bowley's and Kelly's coefficients of skewness are respectively

OPTIONS

- (a) -1 to 1 and -1 to 1
- (b) -3 to 3 and -1 to 1
- (c) -1 to 1 and -3 to 3
- (d) both unbounded

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) -1 to 1 and -1 to 1

EXAM SHORTCUT (AI-generated explanation)

Both are quantile-based measures normalised by their own spread, so both lie between -1 and 1.

32. **FORECAST · FS-364** Combined Mean and Variance**QUESTION**

Group 1 has 30 observations with mean 5 and variance 9; group 2 has 70 with mean 15 and variance 16. What is the combined variance?

OPTIONS

- (a) 34.9
- (b) 25
- (c) 13.9
- (d) 12.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 34.9

EXAM SHORTCUT (AI-generated explanation)

The combined mean is 12, and $[30(9 + 49) + 70(16 + 9)]/100 = 3490/100$.

33. **FORECAST · FS-365** Measures of Location**QUESTION**

For any set of positive numbers, the correct ordering of the three classical means is

OPTIONS

- (a) AM at most GM at most HM
- (b) HM at most GM at most AM
- (c) GM at most HM at most AM
- (d) they are always equal

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) HM at most GM at most AM

EXAM SHORTCUT (AI-generated explanation)

The AM-GM-HM inequality holds for every set of positive numbers, with equality only when all are equal.

34. **FORECAST · FS-366** Measures of Dispersion**QUESTION**

Adding a constant c to every observation changes

OPTIONS

- (a) the mean and the variance
- (b) the mean only
- (c) the variance only
- (d) neither

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the mean only

EXAM SHORTCUT (AI-generated explanation)

Location measures shift by c ; every measure of dispersion is unaffected.

35. **FORECAST · FS-372** Kurtosis**QUESTION**

A distribution with β_2 greater than 3 is called

OPTIONS

- (a) platykurtic
- (b) mesokurtic
- (c) leptokurtic
- (d) symmetric

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) leptokurtic

EXAM SHORTCUT (AI-generated explanation)

Leptokurtic means more peaked with heavier tails than the normal.

FORECAST · STATISTICAL METHODS

Non-Parametric Tests

33 forecast problems · 10 concepts

1. **FORECAST · FS-264** Sign Test

QUESTION

Under the null hypothesis, the sign test statistic follows

OPTIONS

- (a) $\text{Binomial}(n, 1/2)$
- (b) $N(0, 1)$
- (c) a χ^2 distribution
- (d) $U(0, 1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\text{Binomial}(n, 1/2)$

EXAM SHORTCUT (AI-generated explanation)

Each difference is equally likely to be positive or negative under the null, so the count of positives is binomial with $p = 1/2$.

2. **FORECAST · FS-265** Sign Test

QUESTION

A sign test with $n = 7$ rejects only when all 7 signs are positive. What is its size?

OPTIONS

- (a) $1/128$
- (b) $1/64$
- (c) $7/128$
- (d) $1/256$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/128$

EXAM SHORTCUT (AI-generated explanation)

$(1/2)^7 = 1/128$.

3. FORECAST · FS-266 Sign Test**QUESTION**

A sign test with $n = 10$ rejects when the number of plus signs is 0, 1, 9 or 10. What is its size approximately?

OPTIONS

- (a) 0.0215
- (b) 0.109
- (c) 0.05
- (d) 0.002

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.0215

EXAM SHORTCUT (AI-generated explanation)

$$(1 + 10 + 10 + 1)/1024 = 22/1024 = 0.0215.$$

4. FORECAST · FS-267 Runs Test**QUESTION**

With $n_1 = n_2 = 10$, what is the expected number of runs?

OPTIONS

- (a) 10
- (b) 11
- (c) 20
- (d) 9

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 11

EXAM SHORTCUT (AI-generated explanation)

$$E(R) = 2n_1n_2/(n_1 + n_2) + 1 = 200/20 + 1 = 11.$$

5. FORECAST · FS-268 Runs Test**QUESTION**

With $n_1 = n_2 = 10$, what is the variance of the number of runs approximately?

OPTIONS

- (a) 4.74
- (b) 5.26
- (c) 10
- (d) 2.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 4.74

EXAM SHORTCUT (AI-generated explanation)

The formula gives $200(180)/(400 \times 19) = 36000/7600 = 4.74$.

6. FORECAST · FS-269 Runs Test**QUESTION**

How many runs are there in the sequence H H T H T T T H H T?

OPTIONS

- (a) 5
- (b) 6
- (c) 7
- (d) 4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 6

EXAM SHORTCUT (AI-generated explanation)

The blocks are HH, T, H, TTT, HH, T - six runs.

7. FORECAST · FS-270 Mann-Whitney Test**QUESTION**

With $X = \{1, 3, 5\}$ and $Y = \{2, 4, 6\}$, what is the Mann-Whitney U counting pairs with an X below a Y?

OPTIONS

- (a) 6
- (b) 3
- (c) 9
- (d) 4.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 6

EXAM SHORTCUT (AI-generated explanation)

Counting: 1 beats three, 3 beats two, 5 beats one, giving 6.

8. FORECAST · FS-271 Mann-Whitney Test**QUESTION**

With $n_1 = 15$ and $n_2 = 20$, what is $\text{Var}(U)$ under the null?

OPTIONS

- (a) 900
- (b) 150
- (c) 1050
- (d) 300

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 900

EXAM SHORTCUT (AI-generated explanation)

$n_1 n_2 (n_1 + n_2 + 1) / 12 = 300(36) / 12 = 900.$

9. FORECAST · FS-272 Mann-Whitney Test**QUESTION**

With $n_1 = 6$ and $n_2 = 8$, what are $E(U)$ and $\text{Var}(U)$?

OPTIONS

- (a) 24 and 60
- (b) 24 and 48
- (c) 48 and 60
- (d) 24 and 30

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 24 and 60

EXAM SHORTCUT (AI-generated explanation)

$$E(U) = 48/2 = 24 \text{ and } \text{Var}(U) = 48(15)/12 = 60.$$

10. FORECAST · FS-273 Wilcoxon Signed-Rank Test**QUESTION**

For the Wilcoxon signed-rank test with $n = 10$, what is the null expectation of the positive rank sum?

OPTIONS

- (a) 27.5
- (b) 55
- (c) 25
- (d) 30

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 27.5

EXAM SHORTCUT (AI-generated explanation)

$$E(T) = n(n+1)/4 = 110/4 = 27.5.$$

11. **FORECAST · FS-274** Wilcoxon Signed-Rank Test**QUESTION**

For the Wilcoxon signed-rank test with $n = 10$, what is the null variance of the positive rank sum?

OPTIONS

- (a) 96.25
- (b) 192.5
- (c) 55
- (d) 82.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 96.25

EXAM SHORTCUT (AI-generated explanation)

$$n(n+1)(2n+1)/24 = 10(11)(21)/24 = 2310/24 = 96.25.$$

12. **FORECAST · FS-275** Kolmogorov-Smirnov Test**QUESTION**

The Kolmogorov-Smirnov one-sample statistic is

OPTIONS

- (a) the sum of (observed minus expected) squared over expected
- (b) the supremum of the absolute difference between the empirical and the hypothesised distribution functions
- (c) the $\int$ the squared difference of the distribution functions
- (d) the maximum rank

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the supremum of the absolute difference between the empirical and the hypothesised distribution functions

EXAM SHORTCUT (AI-generated explanation)

The statistic is the largest vertical gap between the two distribution functions.

13. **FORECAST · FS-276** Kolmogorov-Smirnov Test**QUESTION**

The non-parametric analogue of the χ^2 goodness-of-fit test for continuous data is

OPTIONS

- (a) the sign test
- (b) the Kolmogorov-Smirnov test
- (c) the run test
- (d) the Mann-Whitney test

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the Kolmogorov-Smirnov test

EXAM SHORTCUT (AI-generated explanation)

KS compares the whole distribution function without any grouping into classes.

14. **FORECAST · FS-277** Kolmogorov-Smirnov Test**QUESTION**

For the sample 0.1, 0.4, 0.7 tested against $U(0, 1)$, what is the Kolmogorov-Smirnov statistic?

OPTIONS

- (a) 0.3
- (b) 0.267
- (c) 0.233
- (d) 0.1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.3

EXAM SHORTCUT (AI-generated explanation)

The largest gap is at 0.7 just after the jump: $|1 - 0.7| = 0.3$.

15. **FORECAST · FS-278** Runs Test**QUESTION**

The Wald-Wolfowitz test examines

OPTIONS

- (a) location only
- (b) whether two samples come from identical distributions, using runs on the pooled ordered data
- (c) the randomness of a single sample
- (d) symmetry

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) whether two samples come from identical distributions, using runs on the pooled ordered data

EXAM SHORTCUT (AI-generated explanation)

Pooling, ordering and counting runs of sample labels tests the equality of the two whole distributions.

16. **FORECAST · FS-279** Kruskal-Wallis Test**QUESTION**

Under the null, the Kruskal-Wallis statistic H is approximately

OPTIONS

- (a) χ^2 with k degrees of freedom
- (b) χ^2 with k - 1 degrees of freedom
- (c) F distributed
- (d) standard normal

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) χ^2 with k - 1 degrees of freedom

EXAM SHORTCUT (AI-generated explanation)

With k groups the statistic has k - 1 degrees of freedom, exactly as in a one-way analysis of variance.

17. **FORECAST · FS-280** Kruskal-Wallis Test**QUESTION**

The Kruskal-Wallis test coincides with a one-way analysis of variance when

OPTIONS

- (a) the data are normal
- (b) the observations are replaced by their ranks
- (c) $k = 2$
- (d) the group sizes are equal

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the observations are replaced by their ranks

EXAM SHORTCUT (AI-generated explanation)

Kruskal-Wallis is precisely the analysis of variance applied to the ranks.

18. **FORECAST · FS-281** Friedman Test**QUESTION**

The non-parametric analogue of a randomised block design is

OPTIONS

- (a) Friedman's test
- (b) the Kruskal-Wallis test
- (c) the Kolmogorov-Smirnov test
- (d) the sign test

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) Friedman's test

EXAM SHORTCUT (AI-generated explanation)

Friedman's test ranks within each block and compares the treatment rank totals.

19. **FORECAST · FS-282** Wilcoxon Signed-Rank Test**QUESTION**

Compared with the sign test, the Wilcoxon signed-rank test additionally uses

OPTIONS

- (a) the magnitudes of the differences, through their ranks
- (b) the sample mean
- (c) the sample variance
- (d) nothing extra

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) the magnitudes of the differences, through their ranks

EXAM SHORTCUT (AI-generated explanation)

The signed-rank test ranks the absolute differences, so larger deviations count for more.

20. **FORECAST · FS-283** Mann-Whitney Test**QUESTION**

If W is the rank sum of the first sample, then U equals

OPTIONS

- (a) W minus $n_1(n_1 + 1)/2$
- (b) W
- (c) $W/2$
- (d) n_1n_2 minus W

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) W minus $n_1(n_1 + 1)/2$

EXAM SHORTCUT (AI-generated explanation)

Subtracting the minimum possible rank sum converts W into U .

21. **FORECAST · FS-284** Runs Test**QUESTION**

The runs test tests

OPTIONS

- (a) location
- (b) randomness
- (c) variance
- (d) normality

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) randomness

EXAM SHORTCUT (AI-generated explanation)

Too few or too many runs indicate a departure from randomness in the sequence order.

22. **FORECAST · FS-285** Runs Test**QUESTION**

With $n_1 = n_2 = n$, what is the maximum possible number of runs?

OPTIONS

- (a) $2n$
- (b) $2n - 1$
- (c) $2n + 1$
- (d) n

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $2n$

EXAM SHORTCUT (AI-generated explanation)

Perfect alternation gives $2n$ runs when the two counts are equal.

23. **FORECAST · FS-286** Asymptotic Relative Efficiency**QUESTION**

Under normality, the asymptotic relative efficiency of the Wilcoxon test relative to the t test is

OPTIONS

- (a) $2/\pi$
- (b) $3/\pi$
- (c) 1
- (d) $\pi/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $3/\pi$

EXAM SHORTCUT (AI-generated explanation)

$3/\pi = 0.955$, so Wilcoxon loses less than 5% of the efficiency even in the t test's ideal setting.

24. **FORECAST · FS-287** Asymptotic Relative Efficiency**QUESTION**

Under normality, the asymptotic relative efficiency of the sign test relative to the t test is

OPTIONS

- (a) $2/\pi$
- (b) $3/\pi$
- (c) 1
- (d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $2/\pi$

EXAM SHORTCUT (AI-generated explanation)

$2/\pi = 0.637$, the same as the efficiency of the median relative to the mean.

25. **FORECAST · FS-288** Chi-Square Tests**QUESTION**

A χ^2 goodness-of-fit test fits a Poisson distribution with λ estimated, using 6 classes. The degrees of freedom are

OPTIONS

- (a) 5
- (b) 4
- (c) 6
- (d) 3

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 4

EXAM SHORTCUT (AI-generated explanation)

6 classes minus 1 for the total minus 1 for the estimated λ gives 4.

26. **FORECAST · FS-289** Chi-Square Tests**QUESTION**

A χ^2 test of independence in a 3 by 4 contingency table has degrees of freedom

OPTIONS

- (a) 12
- (b) 6
- (c) 11
- (d) 7

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 6

EXAM SHORTCUT (AI-generated explanation)

$(3 - 1)(4 - 1) = 6$.

27. **FORECAST · FS-290** Chi-Square Tests**QUESTION**

Yates' continuity correction replaces the absolute difference between observed and expected by

OPTIONS

- (a) that difference minus 0.5
- (b) that difference plus 0.5
- (c) half that difference
- (d) the squared difference

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) that difference minus 0.5

EXAM SHORTCUT (AI-generated explanation)

Subtracting 0.5 shrinks each discrepancy and lowers the statistic, correcting for the discreteness.

28. **FORECAST · FS-291** Median Test**QUESTION**

The median test uses

OPTIONS

- (a) a 2 by $2\chi^2$ test on counts above and below the combined median
- (b) a t test
- (c) a runs test
- (d) the sum of ranks

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) a 2 by $2\chi^2$ test on counts above and below the combined median

EXAM SHORTCUT (AI-generated explanation)

Each observation is classified as above or below the pooled median, and the resulting table is tested.

29. **FORECAST · FS-334** Mann-Whitney Test**QUESTION**

For the Mann-Whitney test with $n_1 = n_2 = 3$, what is the null probability that U equals 0?

OPTIONS

- (a) 1/20
- (b) 1/10
- (c) 1/6
- (d) 1/12

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 1/20

EXAM SHORTCUT (AI-generated explanation)

Exactly one of the $\binom{6}{3} = 20$ arrangements puts all three of one sample below all three of the other.

30. **FORECAST · FS-335** Runs Test**QUESTION**

For the runs test with $n_1 = n_2 = 3$, what is the null probability that the number of runs equals 2?

OPTIONS

- (a) 1/10
- (b) 1/20
- (c) 1/5
- (d) 1/4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 1/10

EXAM SHORTCUT (AI-generated explanation)

Two runs means AAABBB or BBBAAA: 2 of the 20 arrangements.

31. **FORECAST · FS-336** Sign Test**QUESTION**

A sign test with $n = 7$ rejects only when all 7 signs are positive. If the true probability of a positive sign is 0.7, what is the power?

OPTIONS

- (a) 0.082
- (b) 0.5
- (c) 0.7
- (d) 0.35

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.082

EXAM SHORTCUT (AI-generated explanation)

$$\text{Power} = 0.7^7 = 0.0824.$$

32. **FORECAST · FS-337** Kolmogorov-Smirnov Test**QUESTION**

For the one-sample Kolmogorov-Smirnov test with $n = 100$, the approximate 5% critical value is

OPTIONS

- (a) 0.136
- (b) 0.05
- (c) 0.196
- (d) 0.096

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.136

EXAM SHORTCUT (AI-generated explanation)

$$1.36/\sqrt{n} = 1.36/10 = 0.136.$$

33. **FORECAST · FS-360** Wilcoxon Signed-Rank Test**QUESTION**

For the Wilcoxon signed-rank test with $n = 5$, rejecting only when the positive rank sum equals 15, what is the size?

OPTIONS

- (a) $1/32$
- (b) $1/16$
- (c) $5/32$
- (d) $1/64$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/32$

EXAM SHORTCUT (AI-generated explanation)

$T = 15$ requires all five signs positive, which has probability $(1/2)^5$.

FORECAST · STATISTICAL METHODS

Order Statistics

34 forecast problems · 8 concepts

1. **FORECAST · FS-292** Densities of Order Statistics

QUESTION

The density of the sample minimum is

OPTIONS

- (a) $nF^{n-1}f$
- (b) $n(1 - F)^{n-1}f$
- (c) F^n
- (d) $(1 - F)^n$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $n(1 - F)^{n-1}f$

EXAM SHORTCUT (AI-generated explanation)

$P(\min > x) = (1 - F)^n$, and differentiating gives $n(1 - F)^{n-1}f$.

2. **FORECAST · FS-293** Densities of Order Statistics

QUESTION

The density of the sample maximum is

OPTIONS

- (a) $nF^{n-1}f$
- (b) $n(1 - F)^{n-1}f$
- (c) nF^n
- (d) f^n

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $nF^{n-1}f$

EXAM SHORTCUT (AI-generated explanation)

$P(\max \leq x) = F^n$, and differentiating gives $nF^{n-1}f$.

3. FORECAST · FS-294 Exponential Order Statistics**QUESTION**

For 5 independent exponential variables with rate 2, the minimum follows

OPTIONS

- (a) an exponential with rate 2
- (b) an exponential with rate 10
- (c) an exponential with rate $2/5$
- (d) $\Gamma(5, 2)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) an exponential with rate 10

EXAM SHORTCUT (AI-generated explanation)

Rates add for the minimum of independent exponentials: $5(2) = 10$.

4. FORECAST · FS-295 Exponential Order Statistics**QUESTION**

For 3 independent $\text{Exp}(1)$ variables, what is the expected value of the maximum?

OPTIONS

- (a) 1
- (b) $11/6$
- (c) 3
- (d) $3/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $11/6$

EXAM SHORTCUT (AI-generated explanation)

The expected maximum is $1 + 1/2 + 1/3 = 11/6$.

5. FORECAST · FS-296 Uniform Order Statistics**QUESTION**

For 4 independent $U(0,1)$ variables, what is the expected value of the maximum?

OPTIONS

- (a) $3/4$
- (b) $4/5$
- (c) $1/2$
- (d) 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $4/5$

EXAM SHORTCUT (AI-generated explanation)

$$E(\max) = n/(n+1) = 4/5.$$

6. FORECAST · FS-297 Uniform Order Statistics**QUESTION**

For n independent $U(0,1)$ variables, what is the expected value of the minimum?

OPTIONS

- (a) $1/n$
- (b) $1/(n+1)$
- (c) $1/(n-1)$
- (d) 0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/(n+1)$

EXAM SHORTCUT (AI-generated explanation)

The minimum is $\text{Beta}(1, n)$ with mean $1/(n+1)$.

7. FORECAST · FS-298 Range and Spacings**QUESTION**

For n independent $U(0, 1)$ variables, the density of the range is

OPTIONS

- (a) $n(n-1)r^{n-2}(1-r)$
- (b) nr^{n-1}
- (c) $(1-r)^n$
- (d) $n(n-1)r(1-r)^{n-2}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $n(n-1)r^{n-2}(1-r)$

EXAM SHORTCUT (AI-generated explanation)

The range follows $\text{Beta}(n-1, 2)$, whose density is $n(n-1)r^{n-2}(1-r)$.

8. FORECAST · FS-299 Range and Spacings**QUESTION**

For 5 independent $U(0, 1)$ variables, what is the expected range?

OPTIONS

- (a) $2/3$
- (b) $4/5$
- (c) $1/2$
- (d) $5/7$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $2/3$

EXAM SHORTCUT (AI-generated explanation)

The range is $\text{Beta}(4, 2)$ with mean $4/6 = 2/3$.

9. FORECAST · FS-300 Uniform Order Statistics**QUESTION**

The r -th order statistic of a sample of size n from $U(0, 1)$ follows

OPTIONS

- (a) $\text{Beta}(r, n - r)$
- (b) $\text{Beta}(r, n - r + 1)$
- (c) $\text{Beta}(n, r)$
- (d) $U(0, 1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\text{Beta}(r, n - r + 1)$

EXAM SHORTCUT (AI-generated explanation)

The two Beta parameters count the observations at or below and strictly above the r -th.

10. FORECAST · FS-301 Uniform Order Statistics**QUESTION**

For 5 independent $U(0, 1)$ variables, what is the variance of the sample median?

OPTIONS

- (a) $1/28$
- (b) $1/20$
- (c) $1/12$
- (d) $1/25$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/28$

EXAM SHORTCUT (AI-generated explanation)

The median is $\text{Beta}(3, 3)$ with variance $9/(36 \times 7) = 1/28$.

11. **FORECAST · FS-302** Uniform Order Statistics**QUESTION**

For 5 independent $U(0, 1)$ variables, the density of the sample median is

OPTIONS

- (a) $30x^2(1-x)^2$
- (b) $20x^2(1-x)^2$
- (c) $5x^4$
- (d) $6 \times (1-x)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $30x^2(1-x)^2$

EXAM SHORTCUT (AI-generated explanation)

Beta(3, 3) has normalising constant $1/B(3, 3) = 30$.

12. **FORECAST · FS-303** Uniform Order Statistics**QUESTION**

For 3 independent $U(0, 1)$ variables, what is $P(\text{the median exceeds } 1/2)$?

OPTIONS

- (a) $1/2$
- (b) $1/4$
- (c) $3/8$
- (d) $1/8$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/2$

EXAM SHORTCUT (AI-generated explanation)

The median is Beta(2, 2), which is symmetric about $1/2$.

13. **FORECAST · FS-304** Order Statistics via the Binomial**QUESTION**

A sample of size 4 comes from the density $2x$ on $(0, 1)$. What is $P(\text{the third order statistic exceeds } 1/2)$?

OPTIONS

- (a) $243/256$
- (b) $189/256$
- (c) $27/64$
- (d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $243/256$

EXAM SHORTCUT (AI-generated explanation)

$F(1/2) = 1/4$, so the answer is $P(\text{at most 2 of 4 are below } 1/2) = (81 + 108 + 54)/256$.

14. **FORECAST · FS-305** Order Statistics via the Binomial**QUESTION**

For a continuous distribution function F , the variable F applied to the r -th order statistic is

OPTIONS

- (a) $U(0, 1)$
- (b) the r -th order statistic of a uniform sample of the same size
- (c) $\text{Beta}(1, n)$
- (d) a constant

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the r -th order statistic of a uniform sample of the same size

EXAM SHORTCUT (AI-generated explanation)

The probability integral transform is monotone, so it maps order statistics to order statistics.

15. **FORECAST · FS-306** Uniform Order Statistics**QUESTION**

For 3 independent $U(10, 11)$ variables, what is the expected value of the minimum?

OPTIONS

- (a) 10.25
- (b) 10.33
- (c) 10.5
- (d) 10.2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 10.25

EXAM SHORTCUT (AI-generated explanation)

The minimum of 3 standard uniforms has mean $1/4$, and the shift adds 10.

16. **FORECAST · FS-307** Uniform Order Statistics**QUESTION**

For n independent $U(0, \theta)$ variables, $P(\text{the minimum exceeds } x)$ equals

OPTIONS

- (a) $(x/\theta)^n$
- (b) $(1 - x/\theta)^n$
- (c) $1 - (x/\theta)^n$
- (d) $n(1 - x/\theta)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $(1 - x/\theta)^n$

EXAM SHORTCUT (AI-generated explanation)

All n observations must exceed x , each with probability $1 - x/\theta$.

17. FORECAST · FS-308 Exponential Order Statistics**QUESTION**

For n independent $\text{Exp}(\lambda)$ variables, the distribution function of the minimum is

OPTIONS

- (a) $1 - e^{-n\lambda x}$
- (b) $(1 - e^{-\lambda x})^n$
- (c) $e^{-n\lambda x}$
- (d) $1 - e^{-\lambda x/n}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1 - e^{-n\lambda x}$

EXAM SHORTCUT (AI-generated explanation)

The minimum is exponential with rate $n\lambda$.

18. FORECAST · FS-309 Asymptotic Distribution of the Median**QUESTION**

For a large sample from $N(\mu, \sigma^2)$, the asymptotic variance of the sample median is

OPTIONS

- (a) σ^2/n
- (b) $\pi\sigma^2/(2n)$
- (c) $2\sigma^2/(\pi n)$
- (d) $\sigma^2/(4n)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\pi\sigma^2/(2n)$

EXAM SHORTCUT (AI-generated explanation)

$1/(4nf(m)^2)$ with $f(m) = 1/(\sigma\sqrt{2\pi})$ gives $\pi\sigma^2/(2n)$.

19. FORECAST · FS-310 Asymptotic Distribution of the Median**QUESTION**

In general, the asymptotic variance of the sample median is

OPTIONS

- (a) $1/(4nf(m)^2)$
- (b) $1/(nf(m))$
- (c) $f(m)^2/(4n)$
- (d) $1/(2nf(m))$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/(4nf(m)^2)$

EXAM SHORTCUT (AI-generated explanation)

The median's precision depends on the density height at the median.

20. FORECAST · FS-311 Discrete Order Statistics**QUESTION**

The minimum of n independent geometric variables with success probability p is geometric with success probability

OPTIONS

- (a) p^n
- (b) $1 - (1 - p)^n$
- (c) a binomial expression
- (d) a Poisson expression

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1 - (1 - p)^n$

EXAM SHORTCUT (AI-generated explanation)

$P(\min > k) = (1 - p)^{nk}$, so the failure probability is $(1 - p)^n$.

21. FORECAST · FS-312 Exponential Order Statistics**QUESTION**

For the shifted exponential density $e^{-(x-\theta)}$ on $x > \theta$ with n observations and M the minimum, the variable $2n(M - \theta)$ follows

OPTIONS

- (a) χ^2 with $2n$ degrees of freedom
- (b) χ^2 with 2 degrees of freedom
- (c) χ^2 with 1 degree of freedom
- (d) an exponential with rate n

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) χ^2 with 2 degrees of freedom

EXAM SHORTCUT (AI-generated explanation)

$M - \theta$ is exponential with rate n , so $n(M - \theta)$ is $\text{Exp}(1)$ and twice it is χ^2 with 2 degrees of freedom.

22. FORECAST · FS-313 Densities of Order Statistics**QUESTION**

The joint density of the minimum and the maximum is

OPTIONS

- (a) $n(n-1)[F(y) - F(x)]^{n-2}f(x)f(y)$
- (b) $n^2 F^{n-1}(1-F)^{n-1}$
- (c) $n!f(x)f(y)$
- (d) $F(y) - F(x)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $n(n-1)[F(y) - F(x)]^{n-2}f(x)f(y)$

EXAM SHORTCUT (AI-generated explanation)

Two observations are fixed at the extremes and the remaining $n - 2$ lie between them.

23. **FORECAST · FS-314** Densities of Order Statistics**QUESTION**

The joint density of all n order statistics is

OPTIONS

- (a) the product of the individual densities
- (b) $n!$ times that product
- (c) n times that product
- (d) $(n - 1)!$ times that product

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $n!$ times that product

EXAM SHORTCUT (AI-generated explanation)

Every one of the $n!$ orderings of the sample maps to the same ordered vector.

24. **FORECAST · FS-315** Normal Order Statistics**QUESTION**

For 2 independent $N(0, \sigma^2)$ variables, what is the expected value of the minimum?

OPTIONS

- (a) $-\sigma/\sqrt{\pi}$
- (b) $-\sigma/\sqrt{2\pi}$
- (c) $-\sigma/2$
- (d) 0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $-\sigma/\sqrt{\pi}$

EXAM SHORTCUT (AI-generated explanation)

$$E(\min) = -E|X_1 - X_2|/2 = -(\sigma\sqrt{2})\sqrt{2/\pi}/2 = -\sigma/\sqrt{\pi}.$$

25. FORECAST · FS-316 Normal Order Statistics**QUESTION**

For 2 independent $N(0, \sigma^2)$ variables, what is the expected range?

OPTIONS

- (a) $\sigma/\sqrt{\pi}$
- (b) $2\sigma/\sqrt{\pi}$
- (c) $\sigma\sqrt{2}$
- (d) σ

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $2\sigma/\sqrt{\pi}$

EXAM SHORTCUT (AI-generated explanation)

The range is $|X_1 - X_2|$, whose mean is $\sigma\sqrt{2}$ times $\sqrt{2/\pi} = 2\sigma/\sqrt{\pi}$.

26. FORECAST · FS-317 Uniform Order Statistics**QUESTION**

For a sample from $U(0, \theta)$, an unbiased estimator based on the sample maximum is

OPTIONS

- (a) the maximum itself
- (b) $(n+1)/n$ times the maximum
- (c) $n/(n+1)$ times the maximum
- (d) twice the maximum

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $(n+1)/n$ times the maximum

EXAM SHORTCUT (AI-generated explanation)

$E(\max) = n\theta/(n+1)$, so multiplying by $(n+1)/n$ removes the bias.

27. FORECAST · FS-318 Exponential Order Statistics**QUESTION**

For 3 independent $\text{Exp}(1)$ variables, what is $P(\text{the minimum exceeds } 1)$?

OPTIONS

- (a) e^{-1}
- (b) e^{-3}
- (c) $1 - e^{-3}$
- (d) $3e^{-1}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) e^{-3}

EXAM SHORTCUT (AI-generated explanation)

The minimum is $\text{Exp}(3)$, so $P(\min > 1) = e^{-3}$.

28. FORECAST · FS-319 Range and Spacings**QUESTION**

For an $\text{Exp}(\lambda)$ sample, the spacing between the k -th and the $(k + 1) - \text{th}$ order statistics follows

OPTIONS

- (a) an exponential with rate λ
- (b) an exponential with rate $(n - k)\lambda$
- (c) an exponential with rate $k\lambda$
- (d) $\Gamma(k, \lambda)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) an exponential with rate $(n - k)\lambda$

EXAM SHORTCUT (AI-generated explanation)

After the k -th order statistic, $n - k$ observations remain, each restarting by memorylessness.

29. **FORECAST · FS-338** Uniform Order Statistics**QUESTION**

For 3 independent $U(0, 1)$ variables, what is $\text{Cov}(X_1, X_3)$?

OPTIONS

- (a) $1/80$
- (b) $1/16$
- (c) $1/20$
- (d) $1/12$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/80$

EXAM SHORTCUT (AI-generated explanation)

The formula $r(n - s + 1)/((n + 1)^2(n + 2))$ with $r = 1, s = 3$ and $n = 3$ gives $1/80$.

30. **FORECAST · FS-339** Range and Spacings**QUESTION**

For 2 independent $U(0, 1)$ variables, what is the variance of the range?

OPTIONS

- (a) $1/18$
- (b) $1/12$
- (c) $1/24$
- (d) $1/36$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/18$

EXAM SHORTCUT (AI-generated explanation)

The range is $\text{Beta}(1, 2)$ with variance $2/(9 \times 4) = 1/18$.

31. **FORECAST · FS-340** Exponential Order Statistics**QUESTION**

For 3 independent $\text{Exp}(1)$ variables, what is the variance of the maximum?

OPTIONS

- (a) $49/36$
- (b) $11/6$
- (c) 1
- (d) 3

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $49/36$

EXAM SHORTCUT (AI-generated explanation)

The maximum is a sum of independent exponentials with rates 3, 2 and 1, so the variances add: $1/9 + 1/4 + 1$.

32. **FORECAST · FS-341** Uniform Order Statistics**QUESTION**

For 7 independent $U(0, 1)$ variables, what is the variance of the sample median?

OPTIONS

- (a) $1/36$
- (b) $1/28$
- (c) $1/48$
- (d) $1/64$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/36$

EXAM SHORTCUT (AI-generated explanation)

The median is $\text{Beta}(4, 4)$ with variance $16/(64 \times 9) = 1/36$.

33. FORECAST · FS-342 Normal Order Statistics**QUESTION**

For 3 independent $N(0, \sigma^2)$ variables, the expected value of the maximum is

OPTIONS

- (a) $3\sigma/(2\sqrt{\pi})$
- (b) $\sigma/\sqrt{\pi}$
- (c) $\sigma\sqrt{2/\pi}$
- (d) $2\sigma/\sqrt{\pi}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $3\sigma/(2\sqrt{\pi})$

EXAM SHORTCUT (AI-generated explanation)

The classical constant for $n = 3$ gives $E(\max) = 3\sigma/(2\sqrt{\pi}) = 0.846\sigma$.

34. FORECAST · FS-374 Range and Spacings**QUESTION**

In statistical quality control, the sample range is preferred to the sample standard deviation for small subgroups because

OPTIONS

- (a) it is more efficient
- (b) it is far easier to compute and nearly as efficient for subgroups of size 5 or less
- (c) it is unbiased for σ
- (d) it is unaffected by outliers

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) it is far easier to compute and nearly as efficient for subgroups of size 5 or less

EXAM SHORTCUT (AI-generated explanation)

For very small subgroups the range loses little efficiency and is trivial to compute on the shop floor.

FORECAST · STATISTICAL METHODS

Regression Analysis

20 forecast problems · 5 concepts

1. FORECAST · FS-104 Regression Coefficients**QUESTION**

The product of the two regression coefficients b_{yx} and b_{xy} equals

OPTIONS

- (a) r
- (b) r^2
- (c) 1
- (d) σ_y/σ_x

ANSWER (AI-generated forecast item — no official UPSC key exists)**(b)** r^2 **EXAM SHORTCUT** (AI-generated explanation)

$$b_{yx}b_{xy} = (r\sigma_y/\sigma_x)(r\sigma_x/\sigma_y) = r^2.$$

2. FORECAST · FS-105 Regression Lines**QUESTION**

The two regression lines always intersect at

OPTIONS

- (a) the origin
- (b) the point of means
- (c) the point of medians
- (d) the point $(0, \bar{y})$

ANSWER (AI-generated forecast item — no official UPSC key exists)**(b)** the point of means**EXAM SHORTCUT** (AI-generated explanation)

Both lines pass through $(\bar{x}, \bar{y})$ by construction.

3. FORECAST · FS-106 Regression Coefficients**QUESTION**

The lines $2x + 3y = 6$ and $5x + 7y = 12$ are the two regression lines. What is r ?

OPTIONS

- (a) $-\sqrt{14/15}$
- (b) $-\sqrt{15/14}$
- (c) $\sqrt{14/15}$
- (d) $-14/15$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $-\sqrt{14/15}$

EXAM SHORTCUT (AI-generated explanation)

Taking the first as Y on X gives $-2/3$ and the second as X on Y gives $-7/5$; the product $14/15$ is admissible, so $r = -\sqrt{14/15}$.

4. FORECAST · FS-107 Regression Curves**QUESTION**

The joint density is $f(x, y) = x + y$ on the unit square. What is $E(Y \text{ given } X = x)$?

OPTIONS

- (a) $(3x + 2)/(6x + 3)$
- (b) $(x + 1)/2$
- (c) $x/2$
- (d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $(3x + 2)/(6x + 3)$

EXAM SHORTCUT (AI-generated explanation)

The conditional mean is $(x/2 + 1/3)/(x + 1/2) = (3x + 2)/(6x + 3)$.

5. FORECAST · FS-108 Regression Curves**QUESTION**

For $f(x, y) = 2$ on $0 < x < y < 1$, what is $E(Y \text{ given } X = x)$?

OPTIONS

- (a) x
- (b) $(1 + x)/2$
- (c) $2x$
- (d) $x/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $(1 + x)/2$

EXAM SHORTCUT (AI-generated explanation)

Given $X = x$, Y is uniform on $(x, 1)$, whose mean is $(1 + x)/2$.

6. FORECAST · FS-109 Regression Curves**QUESTION**

For $f(x, y) = 2$ on $0 < x < y < 1$, what is $E(X \text{ given } Y = y)$?

OPTIONS

- (a) $y/2$
- (b) y
- (c) $(1 + y)/2$
- (d) y^2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $y/2$

EXAM SHORTCUT (AI-generated explanation)

Given $Y = y$, X is uniform on $(0, y)$, whose mean is $y/2$.

7. FORECAST · FS-110 Regression Coefficients**QUESTION**

With $n = 5$, the sum of x equal to 10, the sum of y equal to 20, the sum of xy equal to 60 and the sum of x^2 equal to 30, the least-squares line is

OPTIONS

- (a) $y = 2x$
- (b) $y = 2x + 1$
- (c) $y = x + 2$
- (d) $y = 4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $y = 2x$

EXAM SHORTCUT (AI-generated explanation)

$$b = (5(60) - 200)/(5(30) - 100) = 100/50 = 2, \text{ and } a = 4 - 2(2) = 0.$$

8. FORECAST · FS-111 Regression Coefficients**QUESTION**

With $r = 0.6$, $\sigma_x = 2$ and $\sigma_y = 3$, what is b_{yx} ?

OPTIONS

- (a) 0.4
- (b) 0.9
- (c) 0.6
- (d) 1.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0.9

EXAM SHORTCUT (AI-generated explanation)

$$b_{yx} = r\sigma_y/\sigma_x = 0.6(3/2) = 0.9.$$

9. FORECAST · FS-112 Residual Variance**QUESTION**

With $\sigma_y = 5$ and $r = 0.8$, what is the residual variance of Y about the regression line?

OPTIONS

- (a) 9
- (b) 16
- (c) 4
- (d) 25

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 9

EXAM SHORTCUT (AI-generated explanation)

$$\sigma_y^2(1 - r^2) = 25(0.36) = 9.$$

10. FORECAST · FS-113 Residual Variance**QUESTION**

The standard error of estimate in a simple linear regression equals

OPTIONS

- (a) $\sigma_y(1 - r^2)$
- (b) σ_y times the square root of $(1 - r^2)$
- (c) $\sigma_y r$
- (d) σ_y divided by the square root of $(1 - r^2)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) σ_y times the square root of $(1 - r^2)$

EXAM SHORTCUT (AI-generated explanation)

The residual VARIANCE is $\sigma_y^2(1 - r^2)$, so the standard error is its square root.

11. **FORECAST · FS-114** Regression Coefficients**QUESTION**

Which of the following can be a genuine pair of regression coefficients (b_{yx}, b_{xy})?

OPTIONS

- (a) 2 and 0.8
- (b) 1.6 and 0.4
- (c) -1 and 0.5
- (d) 3 and 0.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 1.6 and 0.4

EXAM SHORTCUT (AI-generated explanation)

Their product must be r^2 , hence at most 1, and both must share the sign of r : only $1.6(0.4) = 0.64$ qualifies.

12. **FORECAST · FS-115** Regression Coefficients**QUESTION**

How does the arithmetic mean of the two regression coefficients compare with $|r|$?

OPTIONS

- (a) it is at most $|r|$
- (b) it is at least $|r|$
- (c) it equals $|r|$
- (d) the two are unrelated

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) it is at least $|r|$

EXAM SHORTCUT (AI-generated explanation)

The arithmetic mean of two positive numbers is at least their geometric mean, which is $|r|$ here.

13. FORECAST · FS-116 Regression Coefficients**QUESTION**

With $u = (x - a)/h$ and $v = (y - b)/k$, the regression coefficient b_{yx} equals

OPTIONS

- (a) $(h/k)b_{vu}$
- (b) $(k/h)b_{vu}$
- (c) b_{vu}
- (d) $hk b_{vu}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $(k/h)b_{vu}$

EXAM SHORTCUT (AI-generated explanation)

Changing scale multiplies the slope by the ratio of the response scale to the predictor scale.

14. FORECAST · FS-117 Regression Curves**QUESTION**

For $f(x, y) = e^{-y}$ on $0 < x < y$, what is $E(Y \text{ given } X = x)$?

OPTIONS

- (a) x
- (b) $x + 1$
- (c) $2x$
- (d) 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $x + 1$

EXAM SHORTCUT (AI-generated explanation)

Given $X = x$, the conditional density is $e^{-(y-x)}$ on $y > x$, so the mean is $x + 1$.

15. FORECAST · FS-118 Regression Coefficients**QUESTION**

For a regression forced through the origin, the least-squares slope is

OPTIONS

- (a) the sum of xy divided by the sum of x^2
- (b) the sum of y divided by the sum of x
- (c) the sum of xy divided by the sum of y^2
- (d) $\bar{y}$ divided by $\bar{x}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) the sum of xy divided by the sum of x^2

EXAM SHORTCUT (AI-generated explanation)

Minimising the sum of $(y - bx)^2$ gives $b = \sum(xy) / \sum(x^2)$.

16. FORECAST · FS-119 Residual Variance**QUESTION**

A quadratic is fitted by least squares to exactly three points with distinct abscissae. The residual sum of squares is

OPTIONS

- (a) 0
- (b) 1
- (c) undefined
- (d) dependent on the data

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0

EXAM SHORTCUT (AI-generated explanation)

Three parameters fit three points exactly, so the curve interpolates and the residuals vanish.

17. FORECAST · FS-120 Regression Coefficients**QUESTION**

The correlation coefficient r equals

OPTIONS

- (a) $b_{yx}\sigma_y/\sigma_x$
- (b) $b_{yx}\sigma_x/\sigma_y$
- (c) b_{yx}
- (d) b_{yx} divided by σ_x

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $b_{yx}\sigma_x/\sigma_y$

EXAM SHORTCUT (AI-generated explanation)

Rearranging $b_{yx} = r\sigma_y/\sigma_x$ gives $r = b_{yx}\sigma_x/\sigma_y$.

18. FORECAST · FS-121 Regression Curves**QUESTION**

The conditional density of Y given $X = x$ is Γ with shape 2 and scale $1 + x$. What is $E(Y \text{ given } X = x)$?

OPTIONS

- (a) $1 + x$
- (b) $2(1 + x)$
- (c) $(1 + x)/2$
- (d) 2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $2(1 + x)$

EXAM SHORTCUT (AI-generated explanation)

A Γ with shape 2 and scale c has mean $2c$.

19. FORECAST · FS-329 Angle Between Regression Lines**QUESTION**

With $r = 0.6$ and $\sigma_x = \sigma_y$, the tangent of the angle between the two regression lines is

OPTIONS

- (a) $8/15$
- (b) $4/5$
- (c) $3/5$
- (d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $8/15$

EXAM SHORTCUT (AI-generated explanation)

$$\tan(\theta) = [(1 - r^2)/r][\sigma_x\sigma_y/(\sigma_x^2 + \sigma_y^2)] = (0.64/0.6)(1/2) = 8/15.$$

20. FORECAST · FS-368 Residual Variance**QUESTION**

The coefficient of determination r^2 measures

OPTIONS

- (a) the proportion of the variance of Y explained by the linear regression on X
- (b) the proportion of observations lying on the line
- (c) the slope of the regression line
- (d) the correlation itself

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) the proportion of the variance of Y explained by the linear regression on X

EXAM SHORTCUT (AI-generated explanation)

r^2 is the ratio of the explained sum of squares to the total sum of squares.

FORECAST · STATISTICAL METHODS

Sampling Distributions and Exact Tests

62 forecast problems · 5 concepts

1. FORECAST · FS-212 Independence of Mean and Variance**QUESTION**

The independence of the sample mean and the sample variance characterises

OPTIONS

- (a) any symmetric distribution
- (b) the normal distribution alone
- (c) the exponential distribution
- (d) the uniform distribution

ANSWER (AI-generated forecast item — no official UPSC key exists)**(b)** the normal distribution alone**EXAM SHORTCUT** (AI-generated explanation)

The Geary-Lukacs theorem: independence of $\bar{x}$ and s^2 holds if and only if the population is normal.

2. FORECAST · FS-213 Chi-Square Distribution**QUESTION**

For a normal sample of size n , the quantity $(n - 1)s^2/\sigma^2$ follows

OPTIONS

- (a) χ^2 with n degrees of freedom
- (b) χ^2 with $n - 1$ degrees of freedom
- (c) F with $(n - 1, 1)$ degrees of freedom
- (d) t with $n - 1$ degrees of freedom

ANSWER (AI-generated forecast item — no official UPSC key exists)**(b)** χ^2 with $n - 1$ degrees of freedom**EXAM SHORTCUT** (AI-generated explanation)

Estimating the mean costs one degree of freedom.

3. FORECAST · FS-214 Chi-Square Distribution**QUESTION**

If s^2 is defined with divisor n , then $E(s^2)$ equals

OPTIONS

- (a) σ^2
- (b) $(n-1)\sigma^2/n$
- (c) $n\sigma^2/(n-1)$
- (d) σ^2/n

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $(n-1)\sigma^2/n$

EXAM SHORTCUT (AI-generated explanation)

The divisor n makes the estimator biased downwards by the factor $(n-1)/n$.

4. FORECAST · FS-215 Chi-Square Distribution**QUESTION**

For a normal sample with s^2 using divisor $n-1$, what is $\text{Var}(s^2)$?

OPTIONS

- (a) $2\sigma^4/n$
- (b) $2\sigma^4/(n-1)$
- (c) $\sigma^4/(n-1)$
- (d) $2\sigma^2/(n-1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $2\sigma^4/(n-1)$

EXAM SHORTCUT (AI-generated explanation)

$\text{Var}(\chi^2 \text{ with } k \text{ degrees of freedom}) = 2k$ gives $\text{Var}(s^2) = 2\sigma^4/(n-1)$.

5. FORECAST · FS-216 Constructing t Statistics**QUESTION**

$X_1, \dots, X_4$ are independent $N(0, 1)$. For $C(X_1 + X_2)/\sqrt{X_3^2 + X_4^2}$ to follow t with 2 degrees of freedom, C must be

OPTIONS

- (a) 1
- (b) $\sqrt{2}$
- (c) $1/\sqrt{2}$
- (d) 2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 1

EXAM SHORTCUT (AI-generated explanation)

Standardise the numerator by $\sqrt{2}$ and note that t_2 needs a division by $\sqrt{\chi^2/2}$; the two factors of $\sqrt{2}$ cancel.

6. FORECAST · FS-217 Constructing t Statistics**QUESTION**

$X_1, \dots, X_5$ are independent $N(0, 1)$. For $C(X_1 + X_2)/\sqrt{X_3^2 + X_4^2 + X_5^2}$ to follow t with 3 degrees of freedom, C must be

OPTIONS

- (a) $\sqrt{3/2}$
- (b) $\sqrt{2/3}$
- (c) $\sqrt{3}$
- (d) 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\sqrt{3/2}$

EXAM SHORTCUT (AI-generated explanation)

The numerator needs division by $\sqrt{2}$ and the denominator multiplication by $\sqrt{3}$, giving $C = \sqrt{3}/\sqrt{2}$.

7. FORECAST · FS-218 Constructing t Statistics**QUESTION**

$X_1, \dots, X_6$ are independent $N(0, 1)$. For $C(X_1 - X_2)/\sqrt{X_3^2 + X_4^2 + X_5^2 + X_6^2}$ to follow t with 4 degrees of freedom, C must be

OPTIONS

- (a) $\sqrt{2}$
- (b) 2
- (c) $1/\sqrt{2}$
- (d) $\sqrt{4/3}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\sqrt{2}$

EXAM SHORTCUT (AI-generated explanation)

The numerator needs division by $\sqrt{2}$ and the denominator multiplication by $\sqrt{4} = 2$, so $C = 2/\sqrt{2} = \sqrt{2}$.

8. FORECAST · FS-219 Constructing t Statistics**QUESTION**

X_1, X_2, X_3 are independent $N(0, 1)$. For $CX_1/\sqrt{X_2^2 + X_3^2}$ to follow t with 2 degrees of freedom, C must be

OPTIONS

- (a) 1
- (b) $\sqrt{2}$
- (c) 2
- (d) $1/\sqrt{2}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\sqrt{2}$

EXAM SHORTCUT (AI-generated explanation)

X_1 is already standard, so only the $\sqrt{2}$ from the degrees of freedom is needed.

9. FORECAST · FS-220 t and F Distributions**QUESTION**

If T follows the t distribution with n degrees of freedom, then T^2 follows

OPTIONS

- (a) χ^2 with 1 degree of freedom
- (b) F with $(1, n)$ degrees of freedom
- (c) F with $(n, 1)$ degrees of freedom
- (d) χ^2 with n degrees of freedom

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) F with $(1, n)$ degrees of freedom

EXAM SHORTCUT (AI-generated explanation)

Squaring a t gives the ratio of a χ^2 with 1 degree of freedom to a χ^2 with n , each divided by its own degrees of freedom.

10. FORECAST · FS-221 t and F Distributions**QUESTION**

If X follows F with (m, n) degrees of freedom, then $1/X$ follows

OPTIONS

- (a) F with (m, n)
- (b) F with (n, m)
- (c) the ratio of two chi-squares with n and m
- (d) a t distribution

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) F with (n, m)

EXAM SHORTCUT (AI-generated explanation)

Inverting an F swaps the two degrees of freedom.

11. **FORECAST · FS-222** t and F Distributions**QUESTION**

What is the mean of the F distribution with (5, 10) degrees of freedom?

OPTIONS

- (a) 1
- (b) 1.25
- (c) 0.8
- (d) 1.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 1.25

EXAM SHORTCUT (AI-generated explanation)

$$E(F) = n/(n - 2) = 10/8 = 1.25.$$

12. **FORECAST · FS-223** t and F Distributions**QUESTION**

What is the mode of the F distribution with (5, 10) degrees of freedom?

OPTIONS

- (a) 0.5
- (b) 1
- (c) 0.75
- (d) 0.6

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.5

EXAM SHORTCUT (AI-generated explanation)

$$\text{Mode} = [(m - 2)/m][n/(n + 2)] = (3/5)(10/12) = 0.5.$$

13. **FORECAST · FS-224** t and F Distributions**QUESTION**

X follows F with (4, 6) degrees of freedom. What is $E(X)$ times $E(1/X)$?

OPTIONS

- (a) 1
- (b) 3
- (c) 1.5
- (d) 2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 3

EXAM SHORTCUT (AI-generated explanation)

$E(X) = 6/4 = 1.5$ and $1/X$ is $F(6, 4)$ with mean $4/2 = 2$, so the product is 3.

14. **FORECAST · FS-225** t and F Distributions**QUESTION**

For X following F with (m, n) degrees of freedom, $E(X)$ times $E(1/X)$ equals

OPTIONS

- (a) 1
- (b) $mn/((m-2)(n-2))$
- (c) $(m-2)(n-2)/(mn)$
- (d) $n/(n-2)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $mn/((m-2)(n-2))$

EXAM SHORTCUT (AI-generated explanation)

$E(X) = n/(n-2)$ and $E(1/X) = m/(m-2)$.

15. FORECAST · FS-226 Chi-Square Distribution**QUESTION**

What is the mode of the χ^2 distribution with n degrees of freedom, for n at least 2?

OPTIONS

- (a) n
- (b) $n - 2$
- (c) $n - 1$
- (d) $n + 2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $n - 2$

EXAM SHORTCUT (AI-generated explanation)

The χ^2 is Γ with shape $n/2$ and rate $1/2$, whose mode is $(\text{shape} - 1)/\text{rate} = n - 2$.

16. FORECAST · FS-227 Chi-Square Distribution**QUESTION**

What is the moment generating function of the χ^2 distribution with 10 degrees of freedom?

OPTIONS

- (a) $(1 - 2t)^{-5}$
- (b) $(1 - 2t)^{-10}$
- (c) $(1 - t)^{-5}$
- (d) e^{10t}

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $(1 - 2t)^{-5}$

EXAM SHORTCUT (AI-generated explanation)

The χ^2 with k degrees of freedom has MGF $(1 - 2t)^{-k/2}$.

17. **FORECAST · FS-228** Chi-Square Distribution**QUESTION**

What is the expectation of the reciprocal of a χ^2 variable with 4 degrees of freedom?

OPTIONS

- (a) $1/4$
- (b) $1/2$
- (c) 1
- (d) it does not exist

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/2$

EXAM SHORTCUT (AI-generated explanation)

$E(1/\chi^2 \text{ with } n \text{ degrees of freedom}) = 1/(n - 2) = 1/2.$

18. **FORECAST · FS-229** Chi-Square Distribution**QUESTION**

What is the expectation of the reciprocal of a χ^2 variable with 2 degrees of freedom?

OPTIONS

- (a) $1/2$
- (b) 1
- (c) infinite
- (d) 0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) infinite

EXAM SHORTCUT (AI-generated explanation)

The formula $1/(n - 2)$ fails at $n = 2$; the integral diverges at the origin.

19. **FORECAST · FS-230** Chi-Square Distribution**QUESTION**

For a normal sample of size n , what is the expectation of the reciprocal of S^2 ?

OPTIONS

- (a) $1/\sigma^2$
- (b) $(n-1)/((n-3)\sigma^2)$
- (c) $(n-3)/((n-1)\sigma^2)$
- (d) $n/((n-2)\sigma^2)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $(n-1)/((n-3)\sigma^2)$

EXAM SHORTCUT (AI-generated explanation)

Using $E(1/\chi^2) = 1/(k-2)$ with $k = n-1$ gives $(n-1)/((n-3)\sigma^2)$.

20. **FORECAST · FS-231** Chi-Square Distribution**QUESTION**

For a normal sample, the moment generating function of S^2 with divisor $n-1$ is

OPTIONS

- (a) $(1 - 2\sigma^2 t/(n-1))^{-(n-1)/2}$
- (b) $(1 - 2t)^{-(n-1)/2}$
- (c) $e^{\sigma^2 t}$
- (d) $(1 - \sigma^2 t)^{-n}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $(1 - 2\sigma^2 t/(n-1))^{-(n-1)/2}$

EXAM SHORTCUT (AI-generated explanation)

S^2 is $\sigma^2/(n-1)$ times a χ^2 with $n-1$ degrees of freedom, and scaling replaces t by ct .

21. FORECAST · FS-232 Independence of Mean and Variance**QUESTION**

For $n = 5$ independent $N(0, 1)$ variables with T the sum of squared deviations from the sample mean, what is $E(T^2 \text{ times the square of the sample mean})$?

OPTIONS

- (a) $24/5$
- (b) $16/5$
- (c) 8
- (d) 4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $24/5$

EXAM SHORTCUT (AI-generated explanation)

T is χ^2 with 4 degrees of freedom and independent of the mean: $E(T^2) = 8 + 16 = 24$ and $E(\bar{x}^2) = 1/5$.

22. FORECAST · FS-233 Independence of Mean and Variance**QUESTION**

For $n = 4$ independent $N(0, 1)$ variables with T the sum of squared deviations from the sample mean, what is $E(T^2 \text{ times the square of the sample mean})$?

OPTIONS

- (a) $15/4$
- (b) $9/4$
- (c) 6
- (d) 3

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $15/4$

EXAM SHORTCUT (AI-generated explanation)

T is χ^2 with 3 degrees of freedom: $E(T^2) = 6 + 9 = 15$, and $E(\bar{x}^2) = 1/4$.

23. **FORECAST · FS-234** t and F Distributions**QUESTION**

An F test compares the variances of two samples of sizes 20 and 37. The degrees of freedom are

OPTIONS

- (a) 20 and 37
- (b) 19 and 36
- (c) 36 and 19
- (d) 19 and 37

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 19 and 36

EXAM SHORTCUT (AI-generated explanation)

Each sample loses one degree of freedom for its estimated mean.

24. **FORECAST · FS-235** Choice of Statistic**QUESTION**

With σ known and a small normal sample, testing the mean uses

OPTIONS

- (a) the t distribution
- (b) the Z distribution
- (c) the χ^2 distribution
- (d) the F distribution

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the Z distribution

EXAM SHORTCUT (AI-generated explanation)

A known σ leaves no estimated nuisance parameter, so the statistic is exactly standard normal.

25. **FORECAST · FS-236** Choice of Statistic**QUESTION**

The pooled two-sample t test has degrees of freedom

OPTIONS

- (a) $n_1 + n_2$
- (b) $n_1 + n_2 - 2$
- (c) $n_1 + n_2 - 1$
- (d) the smaller of n_1 and n_2 minus 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $n_1 + n_2 - 2$

EXAM SHORTCUT (AI-generated explanation)

Two means are estimated, so two degrees of freedom are lost.

26. **FORECAST · FS-237** Chi-Square Distribution**QUESTION**

The sum of two independent χ^2 variables with a and b degrees of freedom follows

OPTIONS

- (a) χ^2 with a + b degrees of freedom
- (b) χ^2 with ab degrees of freedom
- (c) F with (a, b) degrees of freedom
- (d) twice a χ^2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) χ^2 with a + b degrees of freedom

EXAM SHORTCUT (AI-generated explanation)

Degrees of freedom add for independent chi-squares, as the MGF product shows.

27. **FORECAST · FS-238** Chi-Square Distribution**QUESTION**

A quadratic form built from a normal sample of size 53 whose residuals satisfy 3 independent linear constraints follows

OPTIONS

- (a) χ^2 with 53 degrees of freedom
- (b) χ^2 with 52
- (c) χ^2 with 50
- (d) χ^2 with 49

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) χ^2 with 50

EXAM SHORTCUT (AI-generated explanation)

Each independent linear constraint removes one degree of freedom: $53 - 3 = 50$.

28. **FORECAST · FS-239** Choice of Statistic**QUESTION**

In a one-sample t test, if s^2 doubles while $\bar{x}$ and n stay fixed, the t statistic becomes

OPTIONS

- (a) half its value
- (b) its value divided by the square root of 2
- (c) twice its value
- (d) its value times the square root of 2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) its value divided by the square root of 2

EXAM SHORTCUT (AI-generated explanation)

t is inversely proportional to s, and doubling s^2 multiplies s by $\sqrt{2}$.

29. **FORECAST · FS-240** Chi-Square Distribution**QUESTION**

If s^2 is defined with divisor n , then ns^2/σ^2 follows

OPTIONS

- (a) χ^2 with n degrees of freedom
- (b) χ^2 with $n - 1$ degrees of freedom
- (c) χ^2 with $n + 1$ degrees of freedom
- (d) the F distribution

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) χ^2 with $n - 1$ degrees of freedom

EXAM SHORTCUT (AI-generated explanation)

ns^2 is the same sum of squared deviations as $(n - 1)S^2$, so the degrees of freedom are still $n - 1$.

30. **FORECAST · FS-241** Chi-Square Distribution**QUESTION**

As n grows, the standardised chi-square (χ^2 minus n) divided by the square root of $2n$ converges to

OPTIONS

- (a) $N(0, 1)$
- (b) $N(0, 2)$
- (c) a t distribution
- (d) χ^2 with 1 degree of freedom

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $N(0, 1)$

EXAM SHORTCUT (AI-generated explanation)

The χ^2 with n degrees of freedom is a sum of n independent terms, so the central limit theorem applies.

31. **FORECAST · FS-242** Chi-Square Distribution**QUESTION**

In Fisher's approximation, the square root of twice a χ^2 with n degrees of freedom is approximately normal with mean the square root of $(2n - 1)$ and variance

OPTIONS

- (a) 1
- (b) 2
- (c) n
- (d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 1

EXAM SHORTCUT (AI-generated explanation)

The variance-stabilising square-root transformation makes the variance approximately 1.

32. **FORECAST · FS-243** Chi-Square Distribution**QUESTION**

What is $P(\chi^2$ with 4 degrees of freedom exceeds 2)?

OPTIONS

- (a) e^{-1}
- (b) $2e^{-1}$
- (c) e^{-2}
- (d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $2e^{-1}$

EXAM SHORTCUT (AI-generated explanation)

For even degrees of freedom the survival function is elementary: $e^{-x/2}(1 + x/2) = e^{-1}(2)$.

33. **FORECAST · FS-244** Chi-Square Distribution**QUESTION**

What is $P(\chi^2 \text{ with 2 degrees of freedom exceeds } 4)$?

OPTIONS

- (a) e^{-2}
- (b) e^{-4}
- (c) $1/2$
- (d) e^{-1}

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) e^{-2}

EXAM SHORTCUT (AI-generated explanation)

The χ^2 with 2 degrees of freedom is exponential with mean 2, so the tail is $e^{-x/2}$.

34. **FORECAST · FS-245** t and F Distributions**QUESTION**

The t distribution with 1 degree of freedom is

OPTIONS

- (a) $N(0, 1)$
- (b) the Cauchy distribution
- (c) the Laplace distribution
- (d) χ^2 with 1 degree of freedom

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the Cauchy distribution

EXAM SHORTCUT (AI-generated explanation)

t with 1 degree of freedom is the ratio of two independent standard normals, which is Cauchy.

35. **FORECAST · FS-246** t and F Distributions**QUESTION**

What is the variance of the t distribution with 5 degrees of freedom?

OPTIONS

- (a) 1
- (b) $5/3$
- (c) $5/4$
- (d) 3

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $5/3$

EXAM SHORTCUT (AI-generated explanation)

$$\text{Var}(t_n) = n/(n-2) = 5/3.$$

36. **FORECAST · FS-247** t and F Distributions**QUESTION**

The mean of the t distribution with n degrees of freedom exists for

OPTIONS

- (a) n at least 1
- (b) n greater than 1
- (c) n greater than 2
- (d) all n

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) n greater than 1

EXAM SHORTCUT (AI-generated explanation)

t_1 is Cauchy and has no mean; from $n = 2$ onwards the mean exists and equals 0.

37. **FORECAST · FS-248** t and F Distributions**QUESTION**

What is β_2 for the t distribution with 6 degrees of freedom?

OPTIONS

- (a) 3
- (b) 6
- (c) 4.5
- (d) 9

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 6

EXAM SHORTCUT (AI-generated explanation)

$$\beta_2 = 3 + 6/(n - 4) = 3 + 3 = 6.$$

38. **FORECAST · FS-249** t and F Distributions**QUESTION**

Z is standard normal and Y is an independent χ^2 with n degrees of freedom. Then $Z^2/(Y/n)$ follows

OPTIONS

- (a) F with $(1, n)$ degrees of freedom
- (b) F with $(n, 1)$ degrees of freedom
- (c) χ^2 with $n + 1$ degrees of freedom
- (d) only the square of a t

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) F with $(1, n)$ degrees of freedom

EXAM SHORTCUT (AI-generated explanation)

Z^2 is χ^2 with 1 degree of freedom, so the ratio is $F(1, n)$.

39. FORECAST · FS-250 Chi-Square Distribution**QUESTION**

X is χ^2 with m degrees of freedom and Y an independent χ^2 with n . Then $X/(X + Y)$ follows

OPTIONS

- (a) $\text{Beta}(m, n)$
- (b) $\text{Beta}(m/2, n/2)$
- (c) F with (m, n)
- (d) $U(0, 1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\text{Beta}(m/2, n/2)$

EXAM SHORTCUT (AI-generated explanation)

Chi-squares are Gammas with shapes $m/2$ and $n/2$, so the ratio to the total is $\text{Beta}(m/2, n/2)$.

40. FORECAST · FS-251 t and F Distributions**QUESTION**

X_1 and X_2 are independent $N(0, \sigma^2)$. Then $(X_1 + X_2)/(X_1 - X_2)$ follows

OPTIONS

- (a) $N(0, 1)$
- (b) the Cauchy distribution
- (c) t with 2 degrees of freedom
- (d) F with (1, 1) degrees of freedom

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the Cauchy distribution

EXAM SHORTCUT (AI-generated explanation)

The sum and the difference are independent normals with equal variances, so their ratio is standard Cauchy.

41. **FORECAST · FS-252** t and F Distributions**QUESTION**

X_1 and X_2 are independent $N(0, \sigma^2)$. Then the square of $(X_1 + X_2)/(X_1 - X_2)$ follows

OPTIONS

- (a) F with (1, 1) degrees of freedom
- (b) χ^2 with 1 degree of freedom
- (c) F with (2, 2) degrees of freedom
- (d) t with 1 degree of freedom

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) F with (1, 1) degrees of freedom

EXAM SHORTCUT (AI-generated explanation)

The ratio is standard Cauchy, and the square of a Cauchy is $F(1, 1)$.

42. **FORECAST · FS-253** t and F Distributions**QUESTION**

For a sample of n independent $N(0, \sigma^2)$ variables, n times the square of the sample mean divided by s^2 follows

OPTIONS

- (a) F with $(1, n - 1)$ degrees of freedom
- (b) F with $(n - 1, 1)$ degrees of freedom
- (c) a t distribution
- (d) χ^2 with 1 degree of freedom

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) F with $(1, n - 1)$ degrees of freedom

EXAM SHORTCUT (AI-generated explanation)

The numerator is χ^2 with 1 degree of freedom and the denominator an independent χ^2 with $n - 1$, each scaled.

43. **FORECAST · FS-254** Independence of Mean and Variance**QUESTION**

For independent $N(0, 1)$ variables, what is the expectation of $(X_1 + X_2)$ divided by $(X_3^2 + X_4^2 + X_5^2)$?

OPTIONS

- (a) 0
- (b) 2
- (c) 1
- (d) undefined

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0

EXAM SHORTCUT (AI-generated explanation)

The numerator has mean 0 and the denominator's reciprocal has finite mean, so independence gives 0.

44. **FORECAST · FS-255** Choice of Statistic**QUESTION**

A sample of 25 comes from $N(\mu, 100)$. What is $P(|\bar{x} - \mu| < 4)$ approximately?

OPTIONS

- (a) 0.68
- (b) 0.95
- (c) 0.9544
- (d) 0.99

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 0.9544

EXAM SHORTCUT (AI-generated explanation)

SE = $10/5 = 2$, so 4 is exactly 2 standard errors and the probability is 0.9544.

45. **FORECAST · FS-256** Choice of Statistic**QUESTION**

For independent normal samples, the difference of the two sample means follows

OPTIONS

- (a) a normal law with the two variances divided by their own sample sizes and added
- (b) a t distribution
- (c) a normal law with the variances added and divided by the total sample size
- (d) a χ^2 distribution

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) a normal law with the two variances divided by their own sample sizes and added

EXAM SHORTCUT (AI-generated explanation)

Each mean has variance σ_i^2/n_i , and independence makes those variances add.

46. **FORECAST · FS-257** Choice of Statistic**QUESTION**

The homogeneity of several variances is tested by

OPTIONS

- (a) Bartlett's test
- (b) a t test
- (c) the sign test
- (d) the run test

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) Bartlett's test

EXAM SHORTCUT (AI-generated explanation)

Bartlett's test extends the two-sample F test to k samples.

47. FORECAST · FS-258 Choice of Statistic**QUESTION**

Two independent normal samples have clearly unequal variances. The appropriate test for the means is

OPTIONS

- (a) the pooled t test
- (b) Welch's t test
- (c) an F test
- (d) a χ^2 test

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) Welch's t test

EXAM SHORTCUT (AI-generated explanation)

Welch's procedure adjusts the standard error and the degrees of freedom for unequal variances.

48. FORECAST · FS-259 t and F Distributions**QUESTION**

For $n = 2$, the statistic $(\bar{x} - \mu)/(s/\sqrt{n})$ follows

OPTIONS

- (a) $N(0, 1)$
- (b) the Cauchy distribution
- (c) t with 2 degrees of freedom
- (d) χ^2 with 1 degree of freedom

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the Cauchy distribution

EXAM SHORTCUT (AI-generated explanation)

With $n = 2$ the statistic is t with 1 degree of freedom, which is the Cauchy distribution.

49. **FORECAST · FS-260** Independence of Mean and Variance**QUESTION**

Cochran's theorem decomposes the sum of squared deviations from the population mean into an independent χ^2 with $n - 1$ degrees of freedom and

OPTIONS

- (a) χ^2 with 1 degree of freedom
- (b) χ^2 with n
- (c) a standard normal
- (d) an F variable

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) χ^2 with 1 degree of freedom

EXAM SHORTCUT (AI-generated explanation)

The identity splits n degrees of freedom into $n - 1$ for the residuals and 1 for the mean.

50. **FORECAST · FS-261** Independence of Mean and Variance**QUESTION**

For a normal sample, what is $\text{Corr}(\bar{x}, s^2)$?

OPTIONS

- (a) 1
- (b) 0
- (c) $1/n$
- (d) it depends on μ

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0

EXAM SHORTCUT (AI-generated explanation)

The two are independent, so their correlation is 0.

51. **FORECAST · FS-262** t and F Distributions**QUESTION**

The variance of the F distribution with (m, n) degrees of freedom is finite if and only if

OPTIONS

- (a) $n > 2$
- (b) $n > 4$
- (c) $m > 4$
- (d) both m and n exceed 2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $n > 4$

EXAM SHORTCUT (AI-generated explanation)

The F variance requires the denominator degrees of freedom to exceed 4.

52. **FORECAST · FS-263** t and F Distributions**QUESTION**

Consider: 1. the square of a t with 1 degree of freedom is $F(1, 1)$, 2. the square of a standard normal is χ^2 with 1 degree of freedom, 3. a χ^2 with 2 degrees of freedom is exponential. Which are correct?

OPTIONS

- (a) 1 and 2
- (b) 2 and 3
- (c) all three
- (d) 1 only

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) all three

EXAM SHORTCUT (AI-generated explanation)

All three are standard identities linking the exact sampling distributions.

53. FORECAST · FS-325 Constructing t Statistics**QUESTION**

$X_1, \dots, X_6$ are independent $N(0, 1)$. For $C(X_1 + 2X_2)/\sqrt{X_3^2 + X_4^2 + X_5^2 + X_6^2}$ to follow t with 4 degrees of freedom, C must be

OPTIONS

- (a) $2/\sqrt{5}$
- (b) $1/\sqrt{5}$
- (c) $\sqrt{5}/2$
- (d) $\sqrt{2}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $2/\sqrt{5}$

EXAM SHORTCUT (AI-generated explanation)

$\text{Var}(X_1 + 2X_2) = 5$, so the numerator needs division by $\sqrt{5}$, and the 4 degrees of freedom contribute a factor 2.

54. FORECAST · FS-332 Independence of Mean and Variance**QUESTION**

For $n = 6$ independent $N(0, 1)$ variables with T the sum of squared deviations from the sample mean, what is $E(T^3 \text{ times the fourth power of the sample mean})$?

OPTIONS

- (a) $105/4$
- (b) $315/6$
- (c) 26
- (d) $15/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $105/4$

EXAM SHORTCUT (AI-generated explanation)

$E(T^3) = 5(7)(9) = 315$ and $E(\bar{x}^4) = 3/36 = 1/12$, so the product is $315/12 = 26.25 = 105/4$.

55. FORECAST · FS-333 t and F Distributions**QUESTION**

For X following F with $(6, 6)$ degrees of freedom, what is $E(X)$ times $E(1/X)$?

OPTIONS

- (a) $9/4$
- (b) 1
- (c) $3/2$
- (d) 2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $9/4$

EXAM SHORTCUT (AI-generated explanation)

Both means are $6/4 = 3/2$, so the product is $9/4$.

56. FORECAST · FS-353 t and F Distributions**QUESTION**

$X_1, \dots, X_4$ are independent $N(\mu, \sigma^2)$. Then $(X_1 - X_2)$ divided by the absolute value of $(X_3 - X_4)$ follows

OPTIONS

- (a) t with 1 degree of freedom, that is Cauchy
- (b) $N(0, 1)$
- (c) F with $(1, 1)$
- (d) t with 2 degrees of freedom

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) t with 1 degree of freedom, that is Cauchy

EXAM SHORTCUT (AI-generated explanation)

The numerator is $N(0, 2\sigma^2)$ and the denominator is the absolute value of an independent copy, giving a t with 1 degree of freedom.

57. **FORECAST · FS-354** t and F Distributions**QUESTION**

X is χ^2 with 4 degrees of freedom and Y an independent χ^2 with 6. What are $E(X/Y)$ and $E(X/(X + Y))$?

OPTIONS

- (a) 1 and 0.4
- (b) 1 and 0.5
- (c) 0.67 and 0.4
- (d) 1.5 and 0.4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 1 and 0.4

EXAM SHORTCUT (AI-generated explanation)

$E(X)E(1/Y) = 4(1/4) = 1$, and $X/(X + Y)$ is Beta(2, 3) with mean $2/5$.

58. **FORECAST · FS-355** t and F Distributions**QUESTION**

For X following F with (n, n) degrees of freedom, $E(X)$ equals

OPTIONS

- (a) 1
- (b) $n/(n - 2)$, which exceeds 1
- (c) $n/(n + 2)$
- (d) 0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $n/(n - 2)$, which exceeds 1

EXAM SHORTCUT (AI-generated explanation)

Despite the symmetry of the parameters, the mean is $n/(n - 2)$, strictly above 1.

59. **FORECAST · FS-356** t and F Distributions**QUESTION**

What is $P(F \text{ with } (1, 1) \text{ degrees of freedom exceeding } 1)$?

OPTIONS

- (a) $1/2$
- (b) $1/4$
- (c) $1/3$
- (d) $2/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/2$

EXAM SHORTCUT (AI-generated explanation)

X and $1/X$ have the same distribution, so $P(X > 1) = P(X < 1) = 1/2$.

60. **FORECAST · FS-358** Chi-Square Distribution**QUESTION**

What is the coefficient of variation of a χ^2 variable with 8 degrees of freedom?

OPTIONS

- (a) $1/2$
- (b) $1/4$
- (c) $1/8$
- (d) $1/\sqrt{2}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/2$

EXAM SHORTCUT (AI-generated explanation)

$CV = \sqrt{2n}/n = \sqrt{2/n} = \sqrt{0.25} = 1/2$.

61. **FORECAST · FS-369** Choice of Statistic**QUESTION**

A Type I error is

OPTIONS

- (a) rejecting a true null hypothesis
- (b) accepting a false null hypothesis
- (c) the probability of a correct decision
- (d) the same as the power

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) rejecting a true null hypothesis

EXAM SHORTCUT (AI-generated explanation)

A Type I error is a false positive: rejecting a null hypothesis that is in fact true.

62. **FORECAST · FS-373** Choice of Statistic**QUESTION**

The power of a test is

OPTIONS

- (a) the probability of a Type I error
- (b) the probability of rejecting a false null hypothesis
- (c) the significance level
- (d) 1 minus the significance level

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the probability of rejecting a false null hypothesis

EXAM SHORTCUT (AI-generated explanation)

Power is the probability of correctly rejecting when the alternative is true.

FORECAST · STATISTICAL METHODS

Small-Sample Tests

6 forecast problems · 2 concepts

1. FORECAST · FS-206 Choice of Test**QUESTION**

Blood pressure is measured before and after treatment on the same patients, with normally distributed differences. The appropriate test is

OPTIONS

- (a) the paired t test
- (b) the two-sample t test
- (c) the Z test
- (d) the Mann-Whitney test

ANSWER (AI-generated forecast item — no official UPSC key exists)**(a)** the paired t test**EXAM SHORTCUT** (AI-generated explanation)

The two readings are on the same subject, so the design is paired and the analysis works on the differences.

2. FORECAST · FS-207 Choice of Test**QUESTION**

Two independent small samples come from normal populations with a common unknown variance. The appropriate test for equality of means is

OPTIONS

- (a) the paired t test
- (b) the pooled two-sample t test
- (c) a χ^2 test
- (d) an F test

ANSWER (AI-generated forecast item — no official UPSC key exists)**(b)** the pooled two-sample t test**EXAM SHORTCUT** (AI-generated explanation)

A common unknown variance is exactly the setting for the pooled t test.

3. FORECAST · FS-208 Choice of Test**QUESTION**

To test the hypothesis that the variance of a small normal sample equals σ_0^2 , the appropriate statistic is

OPTIONS

- (a) a t statistic
- (b) a χ^2 statistic with $(n - 1)s^2/\sigma_0^2$
- (c) an F statistic
- (d) a Z statistic

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) a χ^2 statistic with $(n - 1)s^2/\sigma_0^2$

EXAM SHORTCUT (AI-generated explanation)

The scaled sample variance follows a χ^2 distribution with $n - 1$ degrees of freedom.

4. FORECAST · FS-209 Choice of Test**QUESTION**

To compare the variances of two independent normal samples, one uses

OPTIONS

- (a) a t test
- (b) a χ^2 test
- (c) an F test
- (d) a Z test

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) an F test

EXAM SHORTCUT (AI-generated explanation)

The ratio of two independent scaled χ^2 variables is F.

5. FORECAST · FS-210 Paired and One-Sample t**QUESTION**

With a mean difference of 2, a difference standard deviation of 4 and $n = 16$, what are the paired t statistic and its degrees of freedom?

OPTIONS

- (a) 2 with 15 degrees of freedom
- (b) 0.5 with 15 degrees of freedom
- (c) 2 with 16 degrees of freedom
- (d) 8 with 15 degrees of freedom

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 2 with 15 degrees of freedom

EXAM SHORTCUT (AI-generated explanation)

$t = 2/(4/4) = 2$ on $n - 1 = 15$ degrees of freedom.

6. FORECAST · FS-211 Paired and One-Sample t**QUESTION**

With $\bar{x} = 52$, $\mu_0 = 50$, $s = 6$ and $n = 9$, what are the t statistic and its degrees of freedom?

OPTIONS

- (a) 1 with 8 degrees of freedom
- (b) 3 with 8 degrees of freedom
- (c) 1 with 9 degrees of freedom
- (d) 0.33

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 1 with 8 degrees of freedom

EXAM SHORTCUT (AI-generated explanation)

$SE = 6/3 = 2$, so $t = 2/2 = 1$ on 8 degrees of freedom.

FORECAST · STATISTICAL METHODS

Standard Errors and Large-Sample Tests

28 forecast problems · 6 concepts

1. FORECAST · FS-182 Standard Errors**QUESTION**

With $\sigma = 10$ and $n = 25$, what is the standard error of the sample mean?

OPTIONS

- (a) 0.4
- (b) 2
- (c) 4
- (d) 0.5

ANSWER (AI-generated forecast item — no official UPSC key exists)**(b)** 2**EXAM SHORTCUT** (AI-generated explanation)

$$SE = \sigma / \sqrt{n} = 10 / 5 = 2.$$

2. FORECAST · FS-183 Standard Errors**QUESTION**

To halve the standard error of the sample mean, the sample size must be

OPTIONS

- (a) doubled
- (b) quadrupled
- (c) halved
- (d) multiplied by the square root of 2

ANSWER (AI-generated forecast item — no official UPSC key exists)**(b)** quadrupled**EXAM SHORTCUT** (AI-generated explanation)

SE varies as $1/\sqrt{n}$, so halving it needs n to be multiplied by 4.

3. FORECAST · FS-184 Standard Errors**QUESTION**

With $p = 0.2$ and $n = 400$, what is the standard error of the sample proportion?

OPTIONS

- (a) 0.02
- (b) 0.04
- (c) 0.01
- (d) 0.2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.02

EXAM SHORTCUT (AI-generated explanation)

$$\sqrt{0.2 \times 0.8/400} = \sqrt{0.0004} = 0.02.$$

4. FORECAST · FS-185 Confidence Intervals**QUESTION**

With $\bar{x} = 50$, $\sigma = 10$ and $n = 100$, what is the 95% confidence interval for μ ?

OPTIONS

- (a) 48.04 to 51.96
- (b) 49 to 51
- (c) 48 to 52
- (d) 47.4 to 52.6

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 48.04 to 51.96

EXAM SHORTCUT (AI-generated explanation)

SE = 1 and the margin is 1.96, giving 50 plus or minus 1.96.

5. FORECAST · FS-186 Sample Size**QUESTION**

With $\sigma = 10$, what sample size gives a 95% margin of error of 2?

OPTIONS

- (a) 96
- (b) 97
- (c) 100
- (d) 68

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 97

EXAM SHORTCUT (AI-generated explanation)

$$n = (1.96 \times 10/2)^2 = 96.04, \text{ which must be rounded UP to 97.}$$

6. FORECAST · FS-187 Two-Sample Tests**QUESTION**

With $\bar{x}_1 = 100$, $\bar{x}_2 = 95$, $\sigma_1 = \sigma_2 = 10$ and $n_1 = n_2 = 50$, what is the Z statistic?

OPTIONS

- (a) 2.5
- (b) 1.77
- (c) 5
- (d) 3.54

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 2.5

EXAM SHORTCUT (AI-generated explanation)

$$SE = \sqrt{100/50 + 100/50} = 2, \text{ so } Z = 5/2 = 2.5.$$

7. FORECAST · FS-188 Two-Sample Tests**QUESTION**

With $p_1 = 0.6$, $p_2 = 0.5$ and $n_1 = n_2 = 100$, what is the pooled Z statistic approximately?

OPTIONS

- (a) 1.42
- (b) 2
- (c) 0.7
- (d) 1.96

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 1.42

EXAM SHORTCUT (AI-generated explanation)

The pooled proportion is 0.55, giving $SE = \sqrt{0.55 \times 0.45 \times 0.02} = 0.0704$ and $Z = 0.1/0.0704$.

8. FORECAST · FS-189 Standard Errors**QUESTION**

The finite population correction may be dropped when 1. n is much smaller than N, 2. the population is infinite, 3. sampling is with replacement. Which apply?

OPTIONS

- (a) 1 only
- (b) 1 and 2
- (c) all three
- (d) 2 and 3

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) all three

EXAM SHORTCUT (AI-generated explanation)

All three make the correction factor equal or very close to 1.

9. FORECAST · FS-190 Standard Errors**QUESTION**

For a large sample from a normal population, the standard error of the sample median is approximately

OPTIONS

- (a) $\sigma/\sqrt{n}$
- (b) $1.2533\sigma/\sqrt{n}$
- (c) $0.6745\sigma/\sqrt{n}$
- (d) $\sigma/\sqrt{2n}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1.2533\sigma/\sqrt{n}$

EXAM SHORTCUT (AI-generated explanation)

The asymptotic variance of the median is $\pi\sigma^2/(2n)$, whose square root is $1.2533\sigma/\sqrt{n}$.

10. FORECAST · FS-191 Standard Errors**QUESTION**

For a large normal sample, the standard error of the sample standard deviation is approximately

OPTIONS

- (a) $\sigma/\sqrt{n}$
- (b) $\sigma/\sqrt{2n}$
- (c) σ^2 times the square root of $(2/n)$
- (d) σ/n

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\sigma/\sqrt{2n}$

EXAM SHORTCUT (AI-generated explanation)

$\text{Var}(s)$ is about $\sigma^2/(2n)$, so $\text{SE}(s) = \sigma/\sqrt{2n}$.

11. **FORECAST · FS-192** Standard Errors**QUESTION**

For a large normal sample, the standard error of the sample variance is approximately

OPTIONS

- (a) σ^2 times the square root of $(2/n)$
- (b) $\sigma^2/\sqrt{n}$
- (c) $2\sigma^2/n$
- (d) $\sigma/\sqrt{2n}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) σ^2 times the square root of $(2/n)$

EXAM SHORTCUT (AI-generated explanation)

$\text{Var}(s^2) = 2\sigma^4/n$, so the standard error is $\sigma^2\sqrt{2/n}$.

12. **FORECAST · FS-193** Sample Size**QUESTION**

Two groups have $\sigma = 5$ each and the difference of the means to be detected is 1. What is the minimum n per group for two-sided 5% significance?

OPTIONS

- (a) 96
- (b) 193
- (c) 385
- (d) 100

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 193

EXAM SHORTCUT (AI-generated explanation)

$\text{SE} = 5\sqrt{2/n}$ and $1/\text{SE} \geq 1.96$ gives $n \geq 192.08$, so 193.

13. **FORECAST · FS-194** Confidence Intervals**QUESTION**

The two-sided critical value of Z at the 1% level is

OPTIONS

- (a) 1.96
- (b) 2.33
- (c) 2.576
- (d) 3

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 2.576

EXAM SHORTCUT (AI-generated explanation)

2.576 leaves 0.5% in each tail.

14. **FORECAST · FS-195** Confidence Intervals**QUESTION**

The width of a confidence interval for the mean is proportional to

OPTIONS

- (a) n
- (b) $1/n$
- (c) $1/\sqrt{n}$
- (d) the square root of n

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $1/\sqrt{n}$

EXAM SHORTCUT (AI-generated explanation)

The width is $2z\sigma/\sqrt{n}$.

15. **FORECAST · FS-196** Confidence Intervals**QUESTION**

With $\hat{p} = 0.4$ and $n = 100$, what is the approximate half-width of a 95% confidence interval for p ?

OPTIONS

- (a) 0.096
- (b) 0.049
- (c) 0.05
- (d) 0.1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.096

EXAM SHORTCUT (AI-generated explanation)

$SE = \sqrt{0.24/100} = 0.049$ and the half-width is $1.96(0.049) = 0.096$.

16. **FORECAST · FS-197** Large-Sample Tests**QUESTION**

With $\mu_0 = 50$, $\bar{x} = 52$, $\sigma = 8$ and $n = 64$, what is the Z statistic?

OPTIONS

- (a) 2
- (b) 1
- (c) 4
- (d) 0.25

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 2

EXAM SHORTCUT (AI-generated explanation)

$SE = 8/8 = 1$, so $Z = (52 - 50)/1 = 2$.

17. **FORECAST · FS-198** Large-Sample Tests**QUESTION**

What is the two-sided p-value corresponding to $Z = 2$?

OPTIONS

- (a) 0.0228
- (b) 0.0456
- (c) 0.05
- (d) 0.0912

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0.0456

EXAM SHORTCUT (AI-generated explanation)

One tail is 0.0228, so the two-sided p-value is 0.0456.

18. **FORECAST · FS-199** Sample Size**QUESTION**

What sample size is needed to estimate a proportion within 0.05 at 95% confidence, using the worst-case p?

OPTIONS

- (a) 385
- (b) 384
- (c) 400
- (d) 271

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 385

EXAM SHORTCUT (AI-generated explanation)

$n = (1.96)^2(0.25)/(0.05)^2 = 384.16$, rounded up to 385.

19. **FORECAST · FS-200** Two-Sample Tests**QUESTION**

With $W_1 = \sigma_1/\sqrt{n_1}$ and $W_2 = \sigma_2/\sqrt{n_2}$, the standard error of the difference of the two sample means is

OPTIONS

- (a) $W_1 + W_2$
- (b) the square root of the sum of their squares
- (c) their product
- (d) their average

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the square root of the sum of their squares

EXAM SHORTCUT (AI-generated explanation)

Variances add for independent samples, so the standard errors combine in quadrature.

20. **FORECAST · FS-201** Power of a Test**QUESTION**

With $\mu_0 = 50$ tested against $\mu = 52$, $\sigma = 8$, $n = 64$ and a one-sided 5% level, the power is about

OPTIONS

- (a) 0.64
- (b) 0.5
- (c) 0.95
- (d) 0.36

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.64

EXAM SHORTCUT (AI-generated explanation)

The critical mean is 51.645 and $P(\bar{x} > 51.645 \text{ when } \mu = 52) = P(Z > -0.355) = 0.64$.

21. **FORECAST · FS-202** Two-Sample Tests**QUESTION**

When both samples come from populations with the same variance σ^2 , the standard error of the difference of the sample means is

OPTIONS

- (a) σ times the square root of $(1/n_1 + 1/n_2)$
- (b) σ divided by $(\sqrt{n_1} + \sqrt{n_2})$
- (c) σ times the square root of $(n_1 + n_2)$
- (d) σ divided by the square root of $(n_1 n_2)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) σ times the square root of $(1/n_1 + 1/n_2)$

EXAM SHORTCUT (AI-generated explanation)

Factor σ out of $\sqrt{\sigma^2/n_1 + \sigma^2/n_2}$.

22. **FORECAST · FS-203** Large-Sample Tests**QUESTION**

For a large-sample test of the hypothesis that σ equals σ_0 , the Z statistic is

OPTIONS

- (a) $(s - \sigma_0)$ times the square root of $(2n)$, divided by σ_0
- (b) $(s - \sigma_0)$ times the square root of n , divided by σ_0
- (c) $(s^2 - \sigma_0^2)/\sigma_0^2$
- (d) $(s - \sigma_0)/\sigma_0$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $(s - \sigma_0)$ times the square root of $(2n)$, divided by σ_0

EXAM SHORTCUT (AI-generated explanation)

Since $SE(s) = \sigma/\sqrt{2n}$, the standardised statistic carries $\sqrt{2n}$.

23. **FORECAST · FS-204** Standard Errors**QUESTION**

The large-sample standard error of the sample correlation coefficient is

OPTIONS

- (a) $(1 - r^2)/\sqrt{n}$
- (b) $(1 - r)/\sqrt{n}$
- (c) $1/\sqrt{n - 3}$
- (d) $r/\sqrt{n}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $(1 - r^2)/\sqrt{n}$

EXAM SHORTCUT (AI-generated explanation)

$SE(r) = (1 - \rho^2)/\sqrt{n}$, estimated by substituting r for ρ .

24. **FORECAST · FS-205** Sample Size**QUESTION**

Two sample means differ by an amount giving $Z = 1.4$ with $n = 50$ per group. What increase in n per group is needed for $Z = 1.96$?

OPTIONS

- (a) 98
- (b) 48
- (c) 50
- (d) 148

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 48

EXAM SHORTCUT (AI-generated explanation)

The required n is $50(1.96/1.4)^2 = 98$, so the INCREASE is 48.

25. **FORECAST · FS-343** Power of a Test**QUESTION**

For a one-sided Z test at the 5% level with power 0.80 and an effect size equal to σ , the required sample size is

OPTIONS

- (a) 7
- (b) 6
- (c) 16
- (d) 25

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 7

EXAM SHORTCUT (AI-generated explanation)

$$n = (1.645 + 0.842)^2 = 6.185, \text{ rounded up to } 7.$$

26. **FORECAST · FS-359** Standard Errors**QUESTION**

For a normal sample of size 50, the approximate coefficient of variation of the sample standard deviation is

OPTIONS

- (a) 0.1
- (b) 0.2
- (c) 0.05
- (d) 0.14

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.1

EXAM SHORTCUT (AI-generated explanation)

$$SE(s) \text{ is about } \sigma/\sqrt{2n}, \text{ so the CV is } 1/\sqrt{2n} = 1/10.$$

27. **FORECAST · FS-361** Sample Size**QUESTION**

Two proportions differ by 0.05 with a pooled proportion of 0.5 and equal sample sizes. What n per group is needed for $Z = 1.96$?

OPTIONS

- (a) 768
- (b) 384
- (c) 1537
- (d) 500

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 768

EXAM SHORTCUT (AI-generated explanation)

$SE = \sqrt{0.5/n}$ and $0.05/SE = 1.96$ gives $0.5/n = 0.000651$, so $n = 768$.

28. **FORECAST · FS-370** Confidence Intervals**QUESTION**

A 95% confidence interval means that

OPTIONS

- (a) the parameter lies in this particular interval with probability 0.95
- (b) intervals constructed by this procedure contain the parameter 95% of the time in repeated sampling
- (c) 95% of the observations lie in the interval
- (d) the estimate is correct 95% of the time

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) intervals constructed by this procedure contain the parameter 95% of the time in repeated sampling

EXAM SHORTCUT (AI-generated explanation)

The confidence statement is about the PROCEDURE across repeated samples, not about one realised interval.